

OBJECTIVE

After completing this unit, the reader will be able to :

- Understand the major role of transformer in the transmission and distribution of a.c. power.
- Explain the construction and working principle of transformer.
- Derive the emf expression of a single-phase transformer.
- Differentiate the behaviour of transformer under load and no-load conditions.
- Analyze the conditions for maximum efficiency of transformer.

7.1 Introduction

Generating plants that generate the bulk of a.c. power are located in some favourably situated stations. The power so generated is then transmitted to various parts of the country. Since transmission of electrical energy at high voltages is economical, therefore some means is required for stepping up the voltage at generating stations and stepping down the same at the particular area where power is to be distributed. Hence the generated energy is transformed a number of times before it is utilized. This transformation of a.c. from one voltage to another is done by an a.c. static device called 'transformer'. The transformer is a static electromagnetic device that transfers electric power from one circuit to another without changing its frequency. In our country, electrical energy is generated at 6.6 KV or 11 KV or 33 KV, stepped up to 132, 220, 400 or 765 KV with the help of a step-up transformer, then stepped down to 66 KV or 33 KV at grid substations, which further step down the voltage to 11 KV for feeding distributing transformers, stepping down the voltage further to 400/230 V for the customer use. Hence, it is of utmost importance to understand the working of transformer.

7.2. Operating Principle

Transformer plays a major part in the transmission and distribution of a.c. power. It basically works on the mutual induction between two circuits linked by a common magnetic field. Transformer consists of a rectangular or circular laminated magnetic core forming the magnetic frame. The primary coil P and the secondary coil S are wound upon the two cores of the frame as shown in Fig. 7.1.

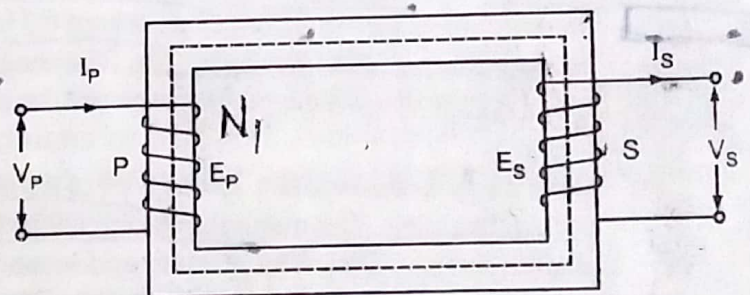


Fig. 7.1. Transformer

The operation of a transformer is based on the principle that energy may be efficiently transferred by induction from one coil to another by means of a varying magnetic flux provided that both the coils are in a common magnetic field. Let the leakage of the magnetic lines of force be negligible so that the two coils P and S are linked by a common magnetic flux ϕ . When an alternating voltage V_p is applied to the primary coil, current I_p flows in it producing magnetic flux in the core (the path shown by dotted line). As the current in the primary winding is alternating, magnetic flux set up is also alternating in nature i.e. its magnitude and direction change with time. As per Faraday's law of electromagnetic induction, a self-induced emf is set up in the primary coil given by :

$$E_p = -N_p \frac{d\phi}{dt}$$

where

N_p = number of turns in primary.

This induced emf is equal and opposite to V_p .

As the flux leakage is negligible, flux ϕ also links with the secondary winding. So, a mutual induced emf is set up in S given by

$$E_s = -N_s \frac{d\phi}{dt}$$

where

N_s = number of turns in secondary.

This mutually induced emf in secondary will circulate the current in the load circuit. Thus energy is transferred from primary winding to the secondary winding by means of electromagnetic induction without any change in frequency. The induced emf in the primary winding opposes the applied voltage and is hence known as back emf.

A step-up transformer transforms a low voltage high current a.c. into a high voltage low current a.c. system. Hence, the secondary contains a large number of turns of fine Cu wire and primary has few turns of thick wire.

A step-down transformer converts a high primary voltage with low current into a low voltage high current. So the number of turns in primary is greater than the number of turns in secondary winding.

7.3. Construction of Transformer

Transformer is a rugged machine requiring minimum amount of repair and maintenance. Owing to the lack of rotating parts, there are no friction or windage losses. Hence, the efficiency of transformer is high. Transformer structure can be divided into four main sections :

(i) **Magnetic Circuit**: Magnetic circuit consists of cores and yokes. The magnetic frame is built of laminated steel. A small percentage of silicon is added to steel to increase its resistivity to eddy currents. The thickness of the laminations is kept 0.3 to 0.5 mm and these laminations are insulated from each other by a thin coating of varnish. The magnetic circuit provides the path for the flow of magnetic flux. Depending on the magnetic circuit, transformers are further classified as (a) core type and (b) shell type.

(a) **Core Type Transformer**: In this type of transformer, the windings surround a considerable part of the core. The magnetic frame consists of two yokes-upper yoke and bottom yoke, as shown in Fig. 7.2(a). The primary and secondary coils are split into two parts, half the turns of primary and half the secondary turns are placed on each core.

(b) **Shell Type Transformer**: In this, the core surrounds a considerable portion of the windings. The magnetic frame consists of a central core and two side cores. Primary and secondary coils are placed on the central core, as shown in Fig. 7.2(b).

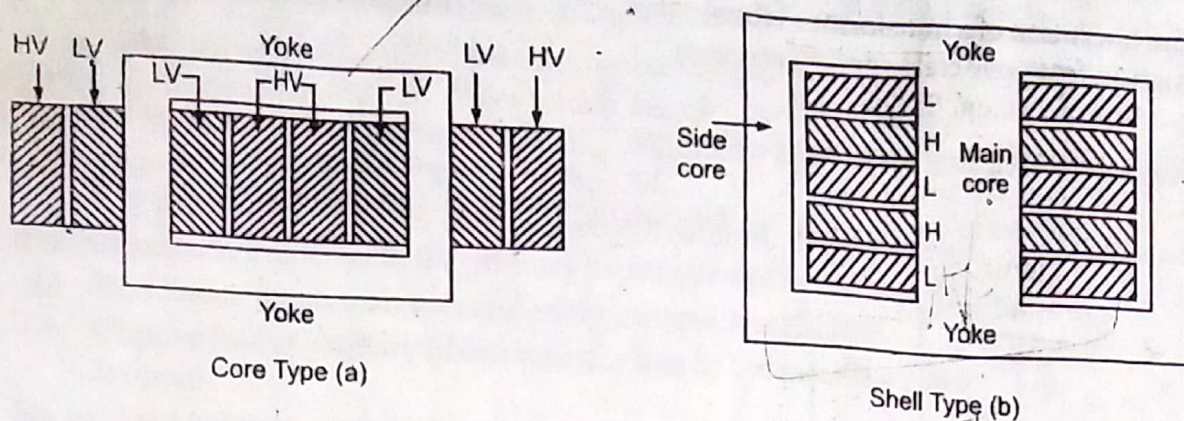


Fig. 7.2.

(ii) **Electrical Circuit** : This mainly consists of primary and secondary windings of copper. Conductors of rectangular cross-section are used for low voltage and high voltage windings for larger transformers. Conductors of circular cross-section are used for high voltage windings in small transformers. Each winding is split up into two equal number of coils and are wrapped on both limbs of the core. By dividing the winding into two halves, there is reduction in leakage flux. The low voltage winding is near the core and HV winding is wrapped over the LV winding. In this way, the amount of high voltage insulation is minimised.

(iii) **Dielectric Circuit** : This consists of insulation. All the conducting parts and magnetic parts of transformer should be well insulated from each other. The insulation should be provided between core and windings and also between low and high voltage windings. Oil ducts are always provided between two solid parts like insulation cylinder and low voltage winding, so as to provide better cooling of these parts.

(iv) **Tank and Accessories** : The assembled transformer with magnetic frame and windings is housed in a proper tank that contains transformer oil. These tanks are made up of steel. A breather consisting of calcium chloride or silica gel is mounted on the transformer tank to extract moisture from the air. A conservator is also provided to take up contraction and expansion of oil without allowing it to come in contact with outside air.

7.4. Ideal Transformer

An ideal transformer consists of two purely inductive coils wound on a loss-free core. Such a transformer has no losses i.e. the primary and secondary windings have no winding resistance, there is no magnetic leakage and no I^2R and core losses.

Consider an ideal transformer whose primary is connected to an alternating voltage V_1 . This voltage V_1 causes an alternating current to flow in primary. Since the primary winding is purely inductive, it draws the magnetising current I_μ . This current I_μ magnetises the core, and lags V_1 by 90° . I_μ produces an alternating flux ϕ which is proportional to current and in-phase with it. This flux is linked with both primary and secondary, so it produces a self induced emf in primary winding which is equal and in opposition to V_1 , also called counter or back emf. Similarly emf E_2 is induced in secondary called mutually induced emf which is anti-phase with V_1 and its magnitude is proportional to the rate of change of flux and number of secondary turns.

EMF Equation :

Let

N_1 = number of primary turns

N_2 = number of secondary turns

ϕ_m = maximum flux in core in webers

f = frequency of a.c. input.

The flux is alternating in nature as shown in Fig. 7.3. It increases from zero value to maximum value ϕ_m in one-quarter of cycle i.e. $1/4f$ seconds.

$$\begin{aligned} \therefore \text{Average rate of change of flux} &= \frac{\phi_m}{1/4f} \\ &= 4f\phi_m \text{ wb/s.} \end{aligned}$$

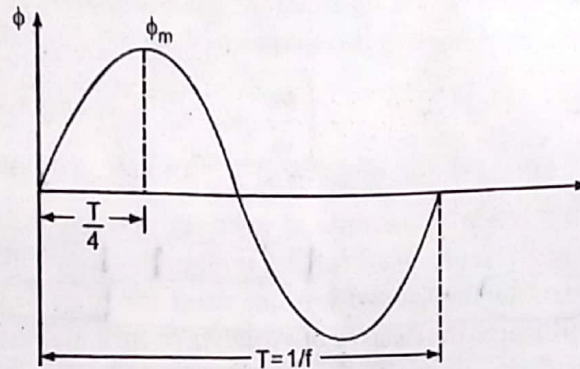


Fig. 7.3. Alternating flux

Rate of change of flux per turn means the induced emf.

$$\therefore \text{Average emf/turn} = 4f\phi_m \text{ volts}$$

If flux ϕ varies sinusoidally, then

$$\text{form factor} = \frac{\text{rms value}}{\text{average value}} = 1.11$$

$$\therefore \text{Rms value of emf/turn} = 1.11 \times 4f\phi_m \text{ volts} = 4.44f\phi_m \text{ volts}$$

So, rms value of induced emf in whole primary winding is

$$\begin{aligned} E_1 &= \text{induced emf per turn} \times \text{no. of primary turns} \\ &= 4.44 f N_1 \phi_m \\ &= 4.44 f N_1 B_m A \end{aligned}$$

and

$$E_2 = 4.44 f N_2 B_m A$$

where A is the cross-sectional area of the core in m^2 and B_m is the maximum flux density.

Also,

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44 f \phi_m = \frac{1}{a}$$

Here a is the transformation ratio. If $N_2 > N_1$, then $a < 1$ and the transformer is step-up. But if $N_2 < N_1$, then $a > 1$ and it is a step-down transformer.

In an ideal transformer, on no-load

$$V_1 = E_1 \text{ and } V_2 = E_2$$

$$\text{Input VA} = \text{output VA}$$

$$V_1 I_1 = V_2 I_2$$

$$\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{1}{a}$$

$\Rightarrow$

7.5. Transformer on No-Load

In an ideal transformer, when primary is connected to alternating voltage V_1 , an alternating current flows which lags V_1 by 90° . The self-induced emf is equal and in opposition to V_1 . Similarly, E_2 is antiphase with V_1 as shown in the phasor diagram of Fig. 7.4.

An ideal transformer has no core and copper losses. But when an actual transformer is on no-load, the primary input current has to supply :

- Iron losses (hysteresis loss and eddy current loss) in core.
- Copper loss in primary (there is no Cu loss in the secondary as it is open).

So, no-load primary current I_0 is not 90° behind V_1 but lags it by an angle less than 90° , as shown in Fig. 7.5.

$$\text{No-load input power } W_0 = V_1 I_0 \cos \phi_0.$$

The primary current I_0 has two components :

- One in phase with V_1 called the active or working or iron loss component I_w . This current I_w supplies the iron loss and small quantity of primary Cu loss. It is given by

$$I_w = I_0 \cos \phi_0.$$

- The second component is in quadrature with V_1 , known as magnetising or reactive or wattless component I_μ , whose function is to produce the magnetic flux. It is given by

$$I_\mu = I_0 \sin \phi_0.$$

Thus I_0 is the vector sum of I_μ and I_w .

$$\text{i.e. } I_0 = \sqrt{I_w^2 + I_\mu^2}.$$

$$\text{Power factor at no-load, } \cos \phi_0 = \frac{I_w}{I_0}.$$

The magnitude of I_w is small as compared to I_0 , so power-factor is very low at no-load. Also the no-load primary current I_0 is very small as compared to full-load primary current (usually 3 to 5% of rated primary current).

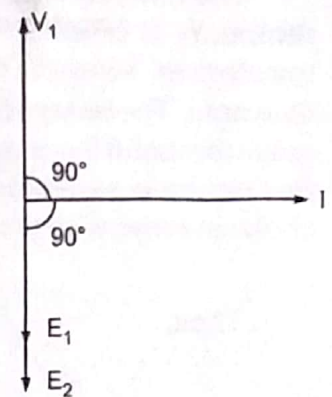


Fig. 7.4. Phasor Diagram of ideal transformer

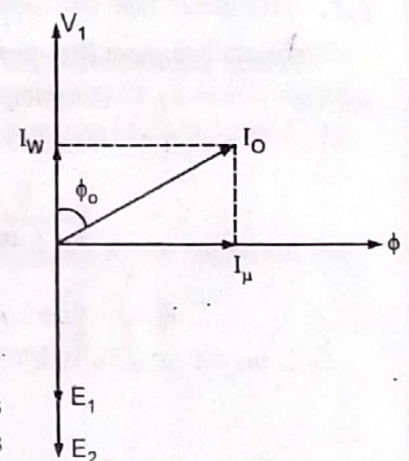


Fig. 7.5. Actual transformer on no-load

7.6. Leakage Reactance

The actual flux set up by the magnetising current in the transformer can be divided into two components :

- Useful flux linking both the windings of transformer which practically remains constant at all values of load.
- Leakage flux linking with only one winding as shown in Fig. 7.6.

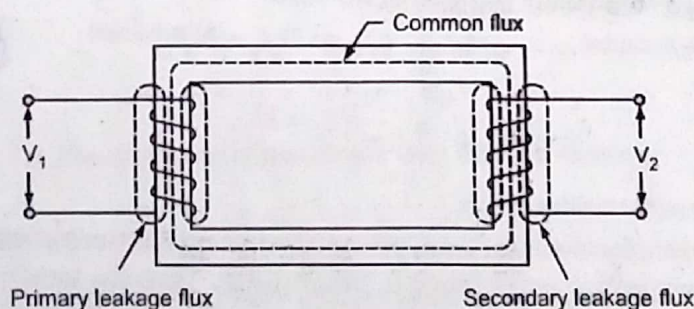


Fig. 7.6. Leakage Reactance of transformer

The flux that links with primary is called primary leakage flux and that which links only with secondary is called secondary leakage flux. The leakage flux is proportional to the load on the transformer. When the transformer is at no-load, leakage flux is negligible. When loaded, leakage flux increases. The leakage flux linking with one winding is due to the total ampere-turns of that winding, while the useful flux is due to the difference between the primary and secondary ampere-turns. Leakage flux produces a self-induced emf in the respective windings. This means that it is equivalent to a small choke in series with the winding.

Thus,

$$X_p = \frac{E_{L1}}{I_1} \text{ and } X_s = \frac{E_{L2}}{I_2}$$

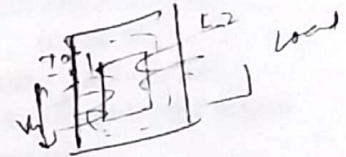
where,

E_{L1} = self-induced emf in primary

E_{L2} = self-induced emf in secondary

X_p = primary leakage reactance

X_s = secondary leakage reactance.



7.7. Transformer on Load

When the secondary of transformer is loaded, as shown in Fig. 7.7, secondary current I_2 is set-up.

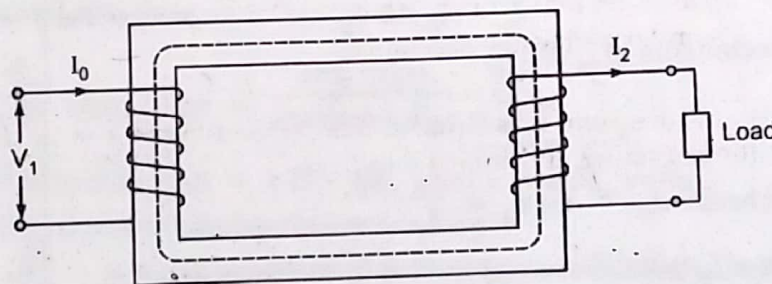


Fig. 7.7. Transformer on Load

The magnitude and phase of I_2 with respect to V_2 depends on the nature of the load. If the load is purely resistive, I_2 is in-phase with V_2 , but if the load is inductive, then I_2 lags V_2 . The secondary current sets up its own flux ϕ_2 which is in opposition to main flux ϕ (set up by I_0). So ϕ_2 weakens ϕ momentarily, hence the back emf E_1 tends to be reduced i.e. for a moment, E_1 becomes less than V_1 and causes more current to flow in primary. Let this additional primary current be I_2' known as the load component of primary current. This current is anti-phase with I_2 and sets up its own flux ϕ_2' which is equal in magnitude but in opposition to ϕ_2 . So the two cancel each other. Hence the net flux passing through the core is approximately same as at no-load.

As

$$\phi_2 = \phi_2', \text{ so } N_2 I_2 = N_1 I_2'$$

$\therefore$

$$I_2' = \frac{N_2}{N_1} I_2 = k I_2$$

$$\phi_2 = \phi_2' \Rightarrow N_2 I_2 = N_1 I_2'$$

$$I_2' = \left(\frac{N_2}{N_1} \right) I_2 = k I_2$$

Here, k is the transformation ratio.

Thus when the transformer is on load, primary winding has two currents—one is I_0 and the other is I_2' which is antiphase with I_2 and k times in magnitude. Thus the total primary current is given by the vector sum of I_0 and I_2' . This is shown in the phasor diagram of Fig. 7.8.

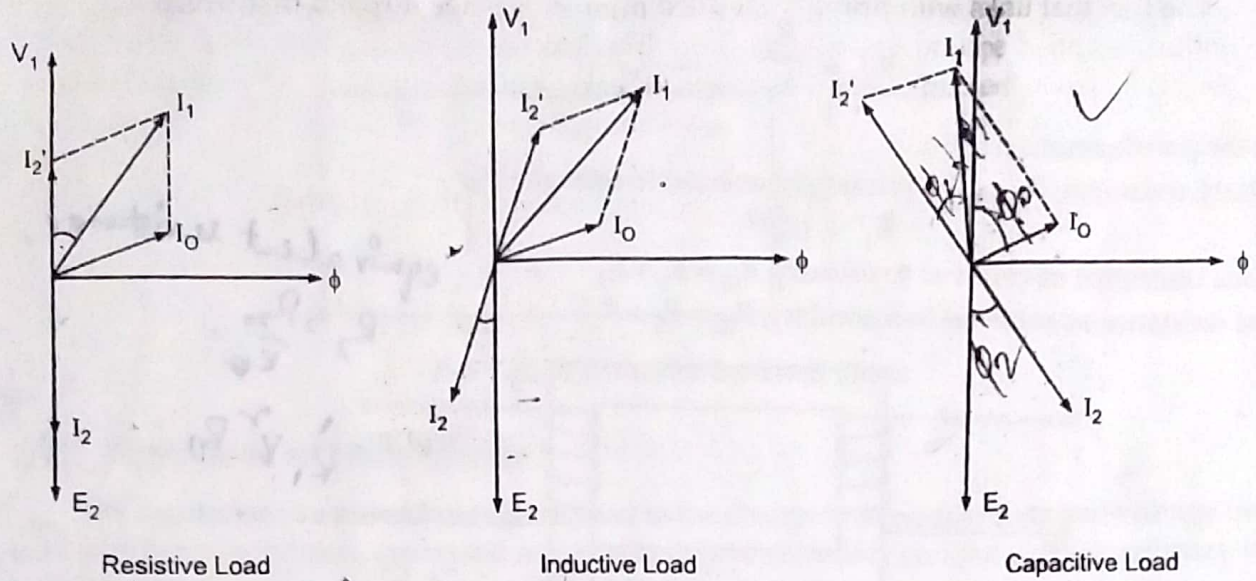


Fig. 7.8. Primary Current

7.8. Transformer with Winding Resistance

An ideal transformer has no winding resistance, but a practical transformer windings (both primary and secondary) possess some resistance. Due to the winding resistance, there is some associated voltage drop also. i.e. the secondary terminal voltage V_2 is less than the induced emf E_2 by $I_2 R_2$, where R_2 is the secondary winding resistance.

$$\therefore V_2 = E_2 - I_2 R_2$$

Also the primary induced emf E_1 is equal to the vector difference of V_1 and $I_1 R_1$ (as shown in the Fig. 7.9). Here R_1 is the resistance of primary winding.

$$\therefore E_1 = V_1 - I_1 R_1$$

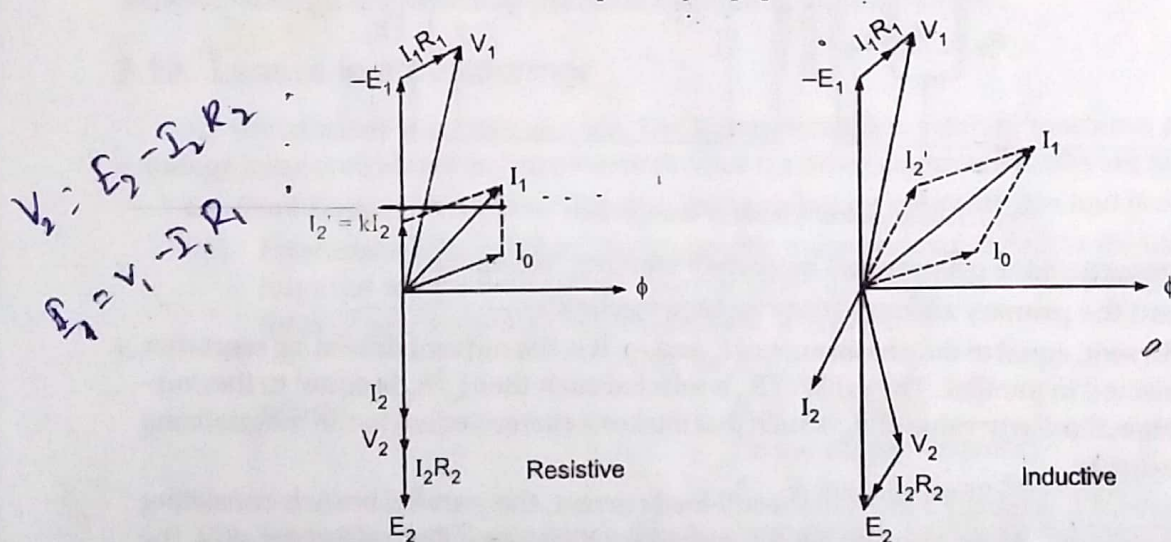


Fig. 7.9. Phasor diagram of transformer with winding resistance

7.9. Equivalent Resistance

Fig. 7.10, shows a transformer with primary and secondary windings having resistances R_1 and R_2 respectively. If R_2' is the equivalent secondary resistance as referred to primary, then

$$R_2' = \frac{R_2}{k^2}$$

where k is the transformation ratio.

Similarly, resistance R_1' of the primary as referred to secondary is

$$R_1' = R_1 k^2$$

∴ Total resistance as referred to primary, $R_{01} = R_1 + R_2'$

Total resistance as referred to secondary, $R_{02} = R_2 + R_1'$

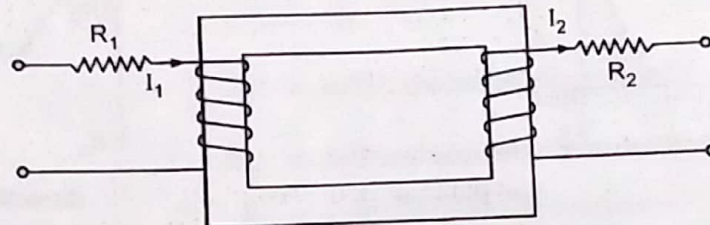
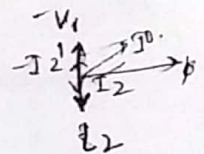


Fig. 7.10. Equivalent Resistance of Transformer

equivalent resistance
 $R_2' = \frac{R_2}{k^2}$
 $R_1' = k^2 R_1$



7.10. Equivalent Circuit of Transformer

The equivalent circuit of a transformer is helpful in determining the behaviour of transformer. The transformer of Fig. 7.10, can be replaced by an equivalent circuit shown in Fig. 7.11.

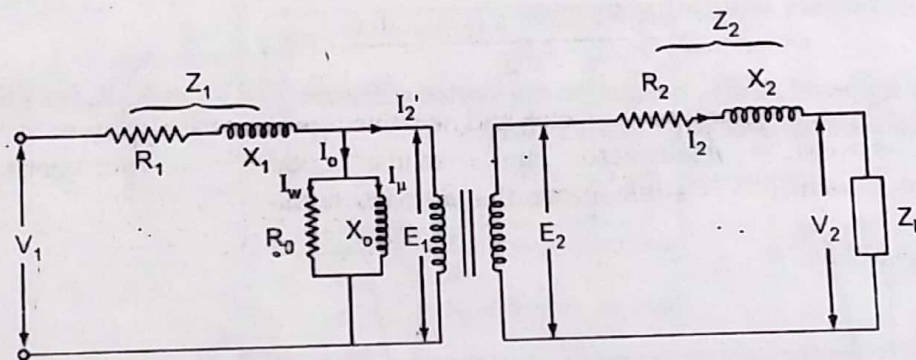


Fig. 7.11. Equivalent Circuit of Transformer

Here, R_1 and R_2 represent the primary and secondary winding resistance.

X_1 and X_2 represent the primary and secondary leakage reactance.

I_0 is the no-load current, equal to the phasor sum of I_w and I_μ . It is the current drawn by reactance X_0 and resistor R_0 connected in parallel. The value of R_0 is selected such that $I_w^2 R_0$ is equal to the core loss of actual transformer. Similarly value of X_0 is such that it takes a current equal to the magnetising current of actual transformer.

As the no-load current I_0 is hardly 3 to 5% of the full-load current, the parallel branch consisting of R_0 and X_0 can be neglected. Now, transferring the secondary resistance R_2 to primary side, the equivalent resistance of primary becomes

$$R_{01} = R_1 + R_2' = R_1 + R_2 k^2.$$

Similarly, the equivalent leakage reactance referred to primary is

$$X_{01} = X_1 + k^2 X_2.$$

Thus the approximate equivalent circuit becomes as shown below in Fig. 7.12.

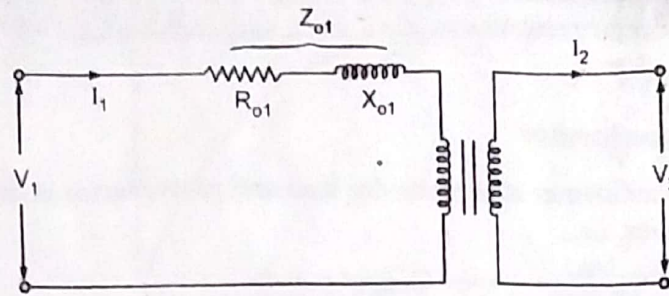


Fig. 7.12. Approximate Equivalent Circuit

7.11. Regulation of Transformer

The regulation of a transformer is defined as the change in secondary terminal voltage from no-load to full-load condition, expressed as a percentage of secondary no-load voltage, primary voltage being constant.

$$\begin{aligned}
 \text{i.e. Percentage regulation} &= \frac{\text{Secondary no-load voltage} - \text{Secondary full-load voltage}}{\text{Secondary no-load voltage}} \times 100 \\
 &= \frac{E_2 - V_2}{E_2} \times 100 \\
 &= \frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{E_2} \times 100
 \end{aligned}$$

The positive sign is for lagging power factor, and negative sign for leading power factor. Regulation should be as small as possible (ideally zero). This is because a good transformer keeps the secondary terminal voltage as constant as possible under all load conditions.

7.12. Losses in a Transformer

As transformer is a static device, the losses present in rotating machines such as friction and windage losses are absent in transformers. But the other associated losses are as follows :

(a) **Core Loss or Iron Loss** : It is due to the pulsating magnetic flux and is of two types :

(i) **Hysteresis Loss** : When the magnetic material is subjected to reversal of flux, power is required for continuous reversal of molecular magnets. This power is dissipated in the form of heat known as hysteresis loss. It is given by

$$W_h = K_h f B_{\max}^{1.6}$$

where

K_h is a constant

f is the supply frequency

$B_{\max}$ is the maximum flux density.

(ii) **Eddy Current Loss** : Since flux is alternating, it links with the magnetic material of core. This induces emf in the core and circulates eddy currents. Power required to maintain these eddy currents, dissipated in the form of heat, is known as eddy current loss and is given by

$$W_e = K_e f^2 B_{\max}^2$$

where t is the thickness of the mass.

(b) **Copper Loss** : These losses occur in primary and secondary windings because of their ohmic resistances.

Total Cu loss = $I_1^2 R_1 + I_2^2 R_2$, where I_1, I_2 are the primary and secondary currents, whereas R_1 and R_2 are the primary and secondary resistances. As I_1 and I_2 vary with load, Cu losses are called variable losses.

7.13. Efficiency of Transformer

The efficiency of a transformer at a particular load and power factor is defined as the ratio of output power to input power. i.e.,

$$\text{Efficiency} = \frac{\text{Output power}}{\text{Input power}}$$

$$= \frac{\text{Output}}{\text{Output} + \text{losses}} = \frac{\text{Output}}{\text{Output} + \text{Copper loss} + \text{Iron loss}}$$

$$\Rightarrow \eta = \frac{\text{Input} - \text{losses}}{\text{Input}} = 1 - \frac{\text{losses}}{\text{Input}}$$

Now

$$\text{Cu loss} = I_1^2 R_1 + I_2^2 R_2$$

$$\text{Iron loss} = \text{hysteresis loss} + \text{eddy current loss}$$

$$= W_h + W_e = W_i$$

$$\text{Input} = V_1 I_1 \cos \phi_1$$

$$\eta = \frac{V_1 I_1 \cos \phi_1 - (I_1^2 R_1 + I_2^2 R_2) - W_i}{V_1 I_1 \cos \phi_1}$$

$$= 1 - \frac{(I_1^2 R_1 + I_2^2 R_2)}{V_1 I_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1}$$

This expression for η can be simplified by expressing Cu loss in terms of primary current and equivalent resistance in terms of primary winding.

$$\therefore \text{Cu loss} = I_1^2 R_{01}$$

and

$$\eta = 1 - \frac{I_1^2 R_{01}}{V_1 I_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1}$$

Differentiating both sides with respect to I_1 ,

Correct this

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For maximum efficiency,

$$\frac{d\eta}{dI_1} = 0$$

$$\therefore \frac{R_{01}}{V_1 \cos \phi_1} = \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

$\Rightarrow$

$$I_1^2 R_{01} = W_i$$

$\Rightarrow$

$$\text{Cu loss} = \text{Iron loss.}$$

So efficiency of a transformer is maximum at a load that makes total Cu loss equal to the constant losses. If we plot efficiency against load current, the curve called efficiency curve is as given below in Fig. 7.13.

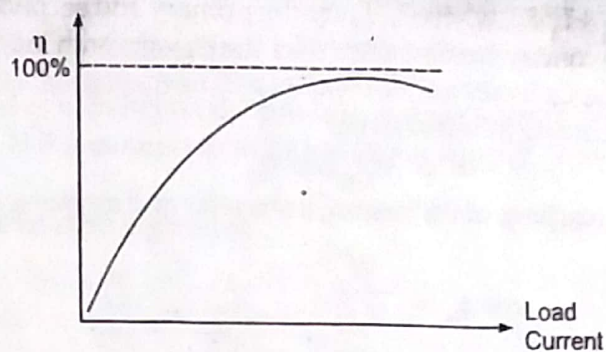


Fig. 7.13. Efficiency Curve

7.14. All Day Efficiency

Commercial efficiency of a transformer is defined as

$$\eta = \frac{\text{Output in watts}}{\text{Input in watts}}$$

But there are some transformers whose performance cannot be judged by this efficiency. For e.g., transformers used for supplying lighting have their primaries energized through-out the day i.e. for 24 hours, but their secondaries supply little or no-load almost the entire day. This implies that core losses occur all-day but Cu loss occurs only when loaded. Hence, such transformers are so designed that the core losses are low, Cu loss being less important. The performance of such a transformer is evaluated on the basis of the energy consumed during a time-period.

$$\eta_{\text{all-day}} = \frac{\text{Output in kWh}}{\text{Input in kWh}} \quad (\text{for 24 hours})$$

This all-day efficiency is always less than the commercial efficiency.

7.15. Transformer Tests

The performance of a transformer can be calculated on the basis of its equivalent circuit which contains four main parameters— R_{01} , X_{01} , core-loss conductance G_0 (or R_0) and magnetizing susceptance B_0 (or X_0). These parameters can be determined by two tests:

- (i) Open-circuit test or Indirect method.
- (ii) Short-circuit test or Impedance method.

7.16. Open-Circuit or No-Load Test

This test is used to determine no-load core loss and no-load I_0 which helps in finding X_0 and R_0 . One winding of transformer is left open (generally high voltage winding) and the other is connected to a supply of normal voltage and frequency as shown in Fig. 7.14.

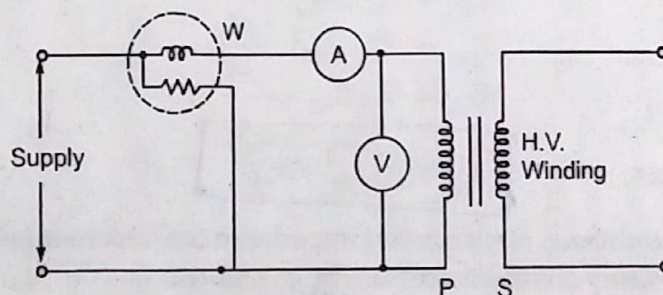


Fig. 7.14. No-Load Test

The ammeter gives the no-load current drawn by the transformer. As no-load current is hardly 3 to 5% of full-load current, the Cu losses under this test will be negligible. So the reading of wattmeter indicates the iron losses occurring in transformer.

Now, $W = V_1 I_0 \cos \phi_0$
where W, V_1, I_0 are the readings of wattmeter, voltmeter and ammeter respectively.

$\therefore$ Power factor, $\cos \phi_0 = \frac{W}{V_1 I_0}$

Magnetising current, $I_\mu = I_0 \sin \phi_0$
and active current $I_w = I_0 \cos \phi_0$

Magnetic reactance, $X_0 = \frac{V_1}{I_\mu}$

and core-loss resistance $R_0 = \frac{V_1}{I_w}$

$W = V I_0 \cos \phi$

$\frac{W}{V I_0} = \cos \phi$

mag current, $I_\mu = I_0 \sin \phi_0$
active current $I_w = I_0 \cos \phi_0$

$X_0 = \frac{V_1}{I_\mu}$

core loss resistance
 $R_0 = \frac{V_1}{I_w}$

7.17. Short-Circuit or Impedance Test

The circuit for the short-circuit test is shown in Fig. 7.15.

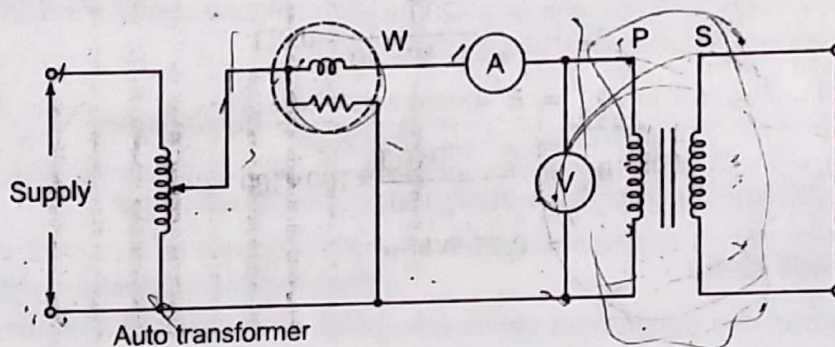


Fig. 7.15. Short Circuit Test

One of the windings of transformer is short-circuited by a conducting wire or an ammeter that will indicate the load current. A low voltage is applied to the other winding. This voltage is slowly increased until full-load current flows in the windings (as indicated by respective ammeters). Normally, the applied voltage is 5 to 7% of the rated voltage of this winding. So flux established in core is small and hence the iron losses are negligible. Thus, the wattmeter reading gives the full-load Cu loss of the transformer. Let W_{SC} , I_{SC} and V_{SC} be the readings of the meters.

Then, $W_{SC} = I_{SC}^2 R_{eq}$

$\Rightarrow R_{eq} = \frac{W_{SC}}{I_{SC}^2}$

Also, $Z_{eq} = \frac{V_{SC}}{I_{SC}}$

and equivalent reactance,

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

So, the equivalent resistance, reactance and impedance can be determined. From these parameters, the regulation and efficiency of transformer can be calculated.

SOLVED EX
Example
are 350 an
no-load

SOLVED EXAMPLES

Example 7.1. The number of turns on the primary and secondary windings of a transformer are 350 and 35 respectively. If it is connected to 2.2 kV, 50 Hz supply, find the secondary voltage at no-load.

Solution. The transformer ratio is given as

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\therefore V_2 = V_1 \times \frac{N_2}{N_1} = 2.2 \times 10^3 \times \frac{35}{350} = 220 \text{ V}$$

Example 7.2. A 3000/220 V, 50 Hz transformer is built on a core having an effective cross-sectional area of 150 cm² and has 80 turns in the low voltage winding. Calculate (i) the maximum flux density in the core and (ii) number of turns in high-voltage winding.

Solution. The emf equation is given by

$$\Rightarrow E_2 = 4.4 f N_2 \phi_m$$

$$200 = 4.44 \times 50 \times 80 \times \phi_m$$

$$\therefore \phi_m = \frac{200}{4.44 \times 50 \times 80} = 0.011$$

Now

$$\phi_m = B_m A$$

$$\therefore B_m = \frac{\phi_m}{A} = \frac{0.011}{150} \times 100 \times 100$$

$$= 0.75 \text{ Wb/m}^2$$

The turns ratio,

$$\frac{N_1}{N_2} = \frac{V_1}{V_2}$$

$\Rightarrow$

$$N_1 = N_2 \times \frac{V_1}{V_2} = \frac{3000}{200} \times 80 = 1200.$$

Example 7.3. A 40 kVA, 3300/240 V, 50 Hz, single-phase transformer has 660 turns on primary. Find the (i) number of turns in secondary, (ii) maximum value of flux in core, (iii) appropriate values of primary and secondary full-load currents. Internal drops in windings are to be ignored.

Solution. We know that

$$(i) \quad \frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$\therefore N_2 = \frac{V_2 N_1}{V_1} = \frac{240}{3300} \times 660 = 48.$$

(ii) As per the emf equation,

$$E_1 = 4.44 N_1 \phi_m$$

Handwritten notes:
 $E = 4.44 f N_2 \phi_m$
 $= 4.44 f N B A$

Output,

Example for 50 kV effective

$$\therefore \theta_m = \frac{E_1}{4.44 N_1 f} = \frac{3300}{4.44 \times 50 \times 660}$$

$$= 0.225 \text{ Wb.}$$

$$(iii) V_1 I_1 = V_2 I_2 = 40k = 40 \times 10^3$$

$$\therefore I_1 = \frac{40 \times 10^3}{V_1} = \frac{40 \times 10^3}{3300} = 12.1 \text{ A}$$

and

$$I_2 = \frac{40 \times 10^3}{V_2} = \frac{40 \times 10^3}{240} = 166.7 \text{ A.}$$

Example 7.4. A 1- ϕ transformer has 500 primary and 1200 secondary turns. The net cross-sectional area of core is 75 cm^2 . If primary winding is connected to a 400 V, 50 Hz supply, calculate the peak value of flux density and the voltage induced in secondary winding.

Solution. Given : $N_1 = 500$, $N_2 = 1200$, $A = 75 \text{ cm}^2$, $E_1 = 400 \text{ V}$, $f = 50 \text{ Hz}$.

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

$$\therefore E_2 = \frac{E_1 N_2}{N_1} = 400 \times \frac{1200}{500} = 960 \text{ V}$$

Now,

$$E_1 = 4.44 f \phi_m N_1$$

$$\therefore \phi_m = \frac{E_1}{4.44 f N_1} = \frac{400}{4.44 \times 50 \times 500}$$

$$= 3.6 \times 10^{-3} \text{ Wb}$$

Flux density,

$$B_m = \phi_m / A$$

$$= \frac{3.6 \times 10^{-3}}{75 \times 10^{-4}} = 0.48 \text{ Wb / m}^2.$$

Example 7.5. A double wound, 1- ϕ transformer is required to step down from 1900 V to 240 V. It is to have 1.5 V per turn. Calculate the required number of turns of primary and secondary windings. The peak value of flux density is required to be not more than 1.2 Wb/m^2 . Calculate the required cross-sectional area of steel core. If output is 10 kVA, calculate the secondary current.

Solution. Transformer winding should have 1.5 V per turn.

$$\therefore N_1 = \frac{1900}{1.5} = 1267$$

$$N_2 = \frac{240}{1.5} = 160$$

It is given that

$$B_m = 1.2 \text{ Wb/m}^2$$

Now,

$$E_1 = 4.44 f N_1 B_m A$$

$$\therefore A = \frac{E_1}{4.44 f N_1 B_m} = \frac{1900}{4.44 \times 50 \times 1267 \times 1.2}$$

Output,

$$= 56.3 \text{ cm}^2$$

$$V_2 I_2 = 10 \text{ kVA},$$

 $\therefore$

$$I_2 = \frac{10 \times 10^3}{240} = 41.7 \text{ A}.$$

Example 7.6. With the instruments connected in high voltage side, the short-circuit test readings for 50 kVA, 2400/240 V transformer are 48 V, 20.8 A and 617 W. Find the leakage impedance, effective resistance and leakage reactance referred to the low voltage side.

Solution. Given that

$$V_{SC} = 48 \text{ V}, I_{SC} = 20.8 \text{ A}$$

 $\therefore$

$$Z_{01} = \frac{V_{SC}}{I_{SC}} = \frac{48}{20.8} = 2.31 \Omega$$

$$\text{Cu loss} = 617 \text{ W} = I_{SC}^2 R_{01}$$

 $\therefore$

$$R_{01} = \frac{617}{I_{SC}^2} = \frac{617}{(20.8)^2} = 1.43 \Omega$$

$$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2} = \sqrt{(2.31)^2 - (1.43)^2}$$

$$= 1.81 \Omega.$$

Transformation ratio,

$$k = \frac{240}{2400} = \frac{1}{10} = 0.1$$

 $\therefore$ Referred to low voltage winding,

Effective resistance,

$$R_{02} = R_{01} \times k^2 = 1.43 \times (0.1)^2$$

$$= 0.0143 \Omega$$

Effective reactance,

$$X_{02} = 1.81 \times 0.1^2$$

$$= 0.0181 \Omega$$

Effective impedance,

$$Z_{02} = 2.31 \times 0.1^2$$

$$= 0.0231 \Omega.$$

Example 7.7. A single-phase 400/110 V transformer takes a no-load current of 4 amp at 0.2 power factor. The secondary supplies a current of 100 amp at a power factor of 0.8 lagging. Determine the (i) current taken by primary winding and (ii) magnetizing reactance and resistance representing core losses.

Solution. Given:

$$\cos \phi = 0.8$$

 $\Rightarrow$

$$\phi = \cos^{-1} 0.8 = 36.87^\circ (\text{lag}).$$

Now,

$$I_2^1 = I_2 \left(\frac{N_2}{N_1} \right) = 100 \times \frac{110}{440} = 25 \text{ amp}.$$

 $\Rightarrow$

$$I_2^1 = 25 \angle -36.87^\circ$$

Given

$$\cos \phi_0 = 0.2$$

 $\Rightarrow$

$$\phi_0 = \cos^{-1} 0.2 = 78.46^\circ (\text{lag})$$

 $\therefore$

$$I_0 = 4 \angle -78.46^\circ$$

Primary current

$$I_1 = I_0 + I_2^1$$

Now,

$$= 4 \angle -78.46^\circ + 25 \angle -36.87^\circ$$

$$= 28.12 \angle 42.29^\circ$$

$$I_w = I_0 \cos \phi_0 = 4 \times 0.2 = 0.8 \text{ amp}$$

$$I_\mu = I_0 \sin \phi_0 = 4 \times 0.98 = 3.92 \text{ amp.}$$

$\therefore$

$$R_w = \frac{V_1}{I_w} = \frac{400}{0.8} = 550 \Omega$$

and

$$X_m = \frac{V_1}{I_\mu} = \frac{400}{3.92} = 112.27 \Omega$$

Example 7.8. The primary and secondary windings of a 50 kVA, 6600/220 V transformer have resistance of 7.8 Ω and 0.0085 Ω respectively. The transformer draws a no-load current of 0.328 A at a power factor of 0.3. Calculate the efficiency at full-load assuming the power factor to be 0.8 lagging.

Solution. $R_1 = 7.8 \Omega$, $R_2 = 0.0085 \Omega$, $I_0 = 0.328 \text{ A}$, $\cos \phi_0 = 0.3$

Full-load current,

$$I_1 = \frac{50 \times 10^3}{6600} = 7.57 \text{ amp.}$$

$$I_2 = \frac{50 \times 10^3}{220} = 227.27 \text{ amp.}$$

$$\begin{aligned} \text{Full-load Cu loss} &= I_1^2 R_1 + I_2^2 R_2 \\ &= (7.57^2 \times 7.8) + (227.27^2 \times 0.0085) \\ &= 886 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Iron loss} &= V_1 I_0 \cos \phi_0 \\ &= 6600 \times 0.328 \times 0.3 \\ &= 649.44 \text{ W.} \end{aligned}$$

For a power factor of 0.8 lagging,

$$\begin{aligned} \text{efficiency } \eta &= \frac{\text{kVA} \times 1000 \times p.f.}{(\text{kVA} \times 1000 \times pf) + W_i + W_{Cu}} \\ &= \frac{50 \times 10^3 \times 0.8}{(50 \times 10^3 \times 0.8) + 649.44 + 886} = 96.3\% \end{aligned}$$

Example 7.9. Calculate the regulation of a transformer in which the ohmic drop is 1% and reactance drop 5% of the voltage at a power factor (i) 0.8 lagging and (ii) 0.8 leading.

$$\text{Solution. Percentage resistive drop} = \frac{I_2 R_{02}}{E_2} \times 100 = 1$$

$$\text{Percentage reactive drop} = \frac{I_2 X_{02}}{E_2} \times 100 = 5$$

(i) When power factor $\cos \phi = 0.8$ lagging, $\sin \phi = 0.6$

$$\begin{aligned}\text{Regulation} &= \frac{I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi}{E_2} \times 100 \\ &= \frac{I_2 R_{02}}{E_2} \times 100 \cos \phi + \frac{I_2 X_{02}}{E_2} \times 100 \sin \phi \\ &= 1 \times 0.8 + 5 \times 0.6 = 3.8\%.\end{aligned}$$

(ii) When power factor $\cos \phi = 0.8$ leading, $\sin \phi = -0.6$

$$\begin{aligned}\therefore \text{Regulation} &= 1 \times 0.8 + 5 \times (-0.6) \\ &= -2.2\%.\end{aligned}$$

Example 7.10. A 200 kVA single-phase transformer has a load of 160 kW at 0.8 pf for 8 hours in a day. For 6 hours, the load is 80 kW at unity pf and for the remaining hours of a day, it runs on no-load. Full-load Cu losses are 3.02 kW and iron losses are 1.6 kW. Find the all-day efficiency.

Solution. Full-load output = 200 kVA

Full-load Cu loss, $P_C = 3.02$ kW

Iron loss, $P_i = 1.6$ kW

$$\begin{aligned}\text{All day output} &= (160 \times 8) + (80 \times 6) \\ &= 1760 \text{ kWh}\end{aligned}$$

$$\text{Iron loss for 24 hours} = 1.6 \times 24 = 38.4 \text{ kWh}$$

$$\begin{aligned}\text{Copper loss for 24 hours} &= \frac{(160/0.8)^2}{200} \times 3.02 \times 8 + \frac{(80/1)^2}{200} \times 3.02 \times 6 \\ &= 27.059 \text{ kWh}\end{aligned}$$

$$\text{Since; output kVA} = \frac{\text{Output kW}}{\text{Power factor}}$$

$$\text{and } Cu \text{ loss} = P_C \left(\frac{\text{Output kVA}}{\text{Rated kVA}} \right)^2$$

$$\begin{aligned}\text{All day input} &= \text{All day output} + \text{iron loss} + \text{Cu loss} \\ &= 1760 + 38.4 + 27.059 \\ &= 1825.46 \text{ kWh}\end{aligned}$$

$$\begin{aligned}\therefore \text{all-day } \eta &= \frac{\text{all-day output}}{\text{all-day input}} \times 100 \\ &= \frac{1760}{1825.46} \times 100 = 96.4\%\end{aligned}$$

SALIENT POINTS

- Transformer is a static device that transfers electrical energy from one circuit to another without a change of frequency.
- If d.c. supply is connected to the transformer primary, it may burn out since there is no self induced emf and heavy current will flow through it.
- The induced emf in the windings is given by :

$$E_1 = 4.44 f N_1 \phi_m$$

and

$$E_2 = 4.44 f N_2 \phi_m$$

where N_1, N_2 are the number of turns and ϕ_m is the flux density.

- An ideal transformer has no losses.
- Voltage transformation ratio is given as

$$k = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

- The primary current, I_0 of a transformer under no-load consists of two parts—in-phase component I_w called the active or iron-loss component and quadrature component I_u called the magnetizing component.
- When a transformer is loaded, primary winding has two currents—one is I_0 and the other is I_2' which is antiphase with I_2 and k times in magnitude.
- The total resistance and reactance referred to primary is given as

$$R_{01} = R_1 + R_2/k^2$$

$$X_{01} = X_1 + X_2/k^2$$

- The total resistance and reactance referred to secondary is

$$R_{02} = k^2 R_1 + R_2$$

$$X_{02} = k^2 X_1 + X_2$$

- Regulation of a transformer should be as small a value as possible. It is given as

$$\% \text{ regulation} = \frac{E_2 - V_2}{E_2} \times 100$$

- Transformer efficiency is given as

$$\eta = \frac{O/P \text{ power}}{I/P \text{ power}} = \frac{\text{Input} - \text{losses}}{\text{Input}}$$

- All-day efficiency is always less than commercial efficiency. It is defined as

$$\eta_{\text{all-day}} = \frac{\text{Output in kWh}}{\text{Input in kWh}} \text{ (for 24 hours)}$$

REVIEW QUESTIONS

A. Conceptual and Conventional Questions

1. What is an ideal transformer? Derive the expression for induced emf in a transformer.
2. Differentiate the following:
 - (i) All-day efficiency and commercial efficiency.
 - (ii) Set-up and step-down transformers.
 - (iii) Short-circuit and open-circuit test.
3. What is meant by regulation of a transformer and how is it affected by the power factor of the load?
4. Show that flux in the core of a transformer is almost constant at all loads.

9

D.C. Machines

OBJECTIVE

After completing this unit, the reader will be able to :

- Understand the operating principle of d.c. generators and motors.
- Identify various parts of a d.c. machine.
- Classify different types of d.c. generators as separately excited and self excited generators.
- Understand the behaviour of various d.c. generators by studying their characteristic curves.
- Classify d.c. motors and analyze their working methodology.
- Explain the behaviour of various d.c. motors through their characteristic curves.

9.1 Introduction

The most popular form of energy (electricity) is not available naturally, but has to be generated from some other form of energy. The process of transformation of electrical energy into mechanical energy or vice-versa is called electromechanical energy conversion. A machine that converts mechanical power to electrical power is called a generator. On the other hand, a d.c. machine that converts electrical power into mechanical power is known as a dc motor. In short, an electric machine is a link joining the electrical and mechanical system.

9.2. Operating Principle of D.C. Generator

The basic principle of working of a d.c. generator is Faraday's law of electromagnetic induction. It states that if there is a relative motion between a conductor placed in a magnetic field and the field, a dynamically induced emf is produced in the conductor. The direction of the induced emf depends on the direction of the magnetic field and the direction of motion and is given by Fleming's right hand rule. Figure 9.1 shows a coil ABCD rotating about its own axis in a uniform magnetic field. When ABCD is parallel to the faces of the field magnets N and S as shown, the flux linking with the loop is maximum, but the rate of change of flux is zero. Thus the induced emf is zero. As the coil continues rotating clockwise, the coil sides begin to cut slowly across the field. The magnitude of the induced emf gradually increases and becomes maximum when the coil is aligned horizontally with the magnets. In the next quarter of

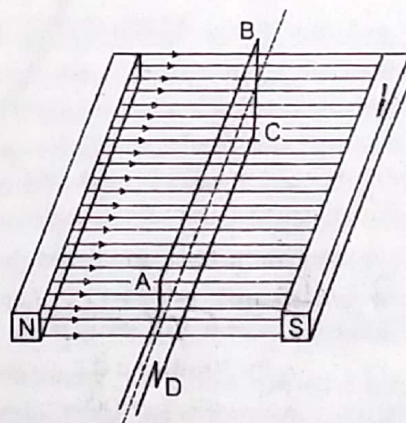


Fig. 9.1. Principle of Operation of D.C. Generator

revolution of the loop (between $\frac{\pi}{2}$ and π radians), the magnitude of the induced emf falls gradually.

Thus the cycle repeats and an alternating emf is generated. This alternating emf is rectified by using split-ring made of a conducting material known as commutator. By employing a large number of coils on the rotating armature of the machine, the resultant current can be made practically constant with respect to time.

9.3. Working Principle of D.C. Motor

The d.c. motor works on the principle that when a current carrying conductor is placed in a magnetic field, mechanical force acts on the conductor due to which the conductor starts rotating in a direction depending on the direction of the current and field, and is given by Fleming's left hand rule. The magnitude of the force experienced by the conductor is given by :

$$F = BIl \text{ newtons}$$

where B is the field strength in Wb/m^2

I is the current flowing through the conductor in amps

l is the length of the conductor in metres.

Figure 9.2 shows a 4-pole dc motor connected to the dc supply. Say, the current in armature conductors is outwards under the N-poles as shown by dots and inwards under the S-poles as shown by crosses. When the dc current passes through the commutator, it is converted into a.c. The direction of the force on each conductor can be determined by Fleming's left hand rule which states that if the fore-finger of the left hand represents the field direction, second finger represents the current direction, then the thumb gives the direction of the motion. Thus each conductor experiences a force that tends to rotate the motor armature in clockwise direction.

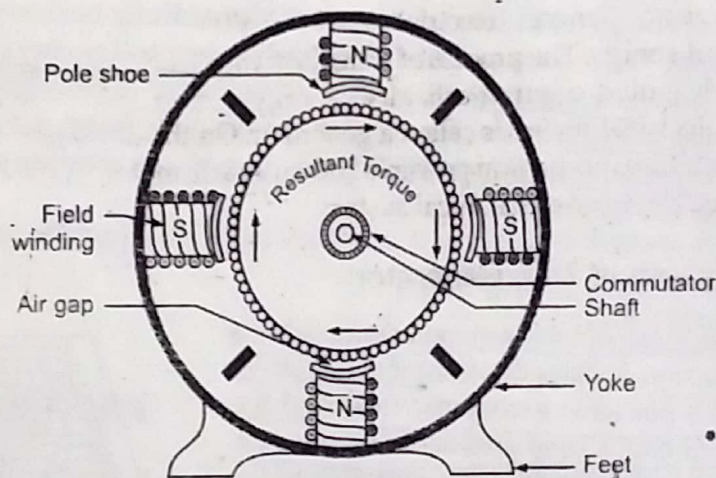


Fig. 9.2. D.C. Motor

9.4. Parts of a D.C. Machine

A d.c. motor and a d.c. generator, both need four main parts for their operation—a stator which is the field system to produce magnetic field, a rotor which is the armature consisting of conductors, an armature to rotate the conductors in the magnetic field and a commutator to convert the a.c. wave to d.c. or to reverse the input wave. The detailed description of each part is given below :

1. **Field system** : It's function is to create a uniform magnetic field. It further consists of four main parts :

(a) **Yoke** : Cast iron or preferably cast steel is used for the yokes. Yoke acts as a frame of the machine and carries the flux produced by the poles. It also provides mechanical support for the entire assembly. Yoke forms a part of the magnetic circuit of the machine.

- (b) **Pole core** : It is used to carry the insulated coils of field current. The exciting current passing through these field coils produce the magnetic field in the whole pole body.
 - (c) **Pole shoes** : They support the field coils placed on the pole body and also spread the total flux over a greater area, thereby reducing the air gap reluctance. The poles are bolted to the frame or yoke, and are always laminated to avoid heating and eddy current losses.
 - (d) **Magnetising coils** : Also called the field coils, they provide the required number of ampere-turns of excitation. These coils are baked and impregnated under vacuum to avoid air gap between conductors.
2. **Armature** : It is the rotating part of the machine, cylindrical in shape, consisting of thin laminations insulated from each other and arranged so that electric currents are induced in these wires when the armature is rotated in the magnetic field. The thin laminations also reduce the eddy current losses. The armature core is provided with slots on the periphery to house the armature coils, as shown in figure 9.3. Holes or ventilating ducts are provided to improve the cooling and to remove the heat generated by the losses taking place in the armature. A small air gap exists between the pole pieces and the armature so that there is no rubbing in the machine.

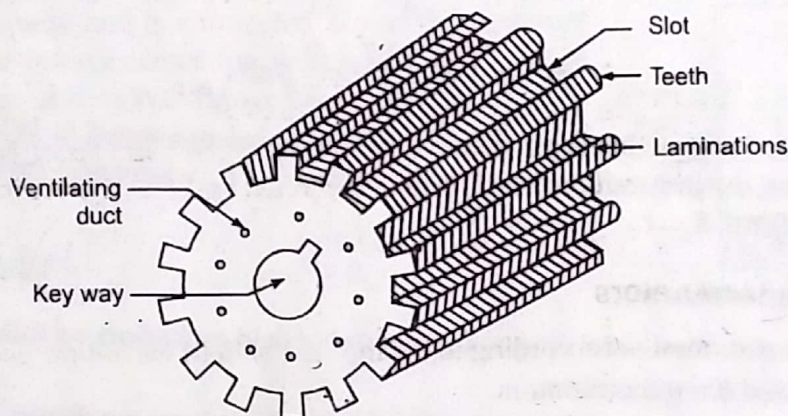


Fig. 9.3. Armature of a D.C. Machine

3. **Commutator** : The armature coils are connected to the commutator which rectifies the alternating emf induced in the coils. The commutator reverses the connections to the external circuit at the instant of each reversal of current in the armature coils. The commutator is cylindrical in shape, built up of wedge shaped segments, which are insulated from each other by thin layers of mica. A V-shaped groove is provided at the end of the segments to hold them together. Finally, the commutator is subjected to severe electrical and mechanical loading.
4. **Brushes** : The function of the brushes is to collect the current from commutator and supply it to external circuit. Generally, brushes are made up of carbon and graphite, and are housed in brush holders. The number of brush holders is equal to the number of main poles in the machine.
5. **Bearings** : They form an important part of a rotating machine. Bearings are used to support the rotating part and allow smooth motion with minimum friction.

9.5. EMF Equation

Let

Z = Total number of armature conductors

P = number of poles

A = number of parallel paths in the armature

N = speed of armature in rpm.

During one revolution of armature, each armature conductor cuts the magnetic flux P times.

So, flux cut by one conductor per revolution = $P\phi$ Wb.

Now, number of revolutions made by armature per minute = N

$$\therefore \text{Number of revolutions per second} = \frac{N}{60}$$

$$\text{So, flux cut by each conductor per second} = P\phi \times \frac{N}{60} \text{ Wb}$$

$$\therefore \text{Average emf induced in one conductor} = P\phi \times \frac{N}{60} \text{ Volts}$$

$$\text{Emf induced in all the } Z \text{ conductors} = \frac{\phi NPZ}{60} \text{ Volts}$$

If the Z conductors are divided into A parallel paths, then emf induced in each path

$$= \frac{1}{A} \times \frac{\phi NPZ}{60} \text{ Volts}$$

$$= \frac{\phi ZN}{60} \times \frac{P}{A} \text{ Volts}$$

Thus, the induced emf is directly proportional to flux per pole ϕ and speed N . When the machine is operating as a generator, the induced emf is called generated emf E_g , whereas in case of a motor, it is called as counter or back emf, E_b .

9.6. Types of D.C. Generators

D.C. generators are classified according to the method of field excitation, as follows :

- (i) Separately excited d.c. generators.
- (ii) Self-excited d.c. generators, which are further classified as : shunt machines, series machines and compound machines.

(i) Separately Excited D.C. Generators : The poles of a d.c. generator are of fixed polarity and thus require energization from a d.c. source. When the field winding is excited from an external source like a battery, the generator is called a separately excited generator. As shown in figure 9.4, the current in the field winding can be varied with the help of a rheostat. If I_a is the armature current, which is same as the load current I_L , then the terminal voltage V is given as :

$$V = E_g - I_a R_a$$

Where R_a is the armature resistance,
and E_g is the generated emf.

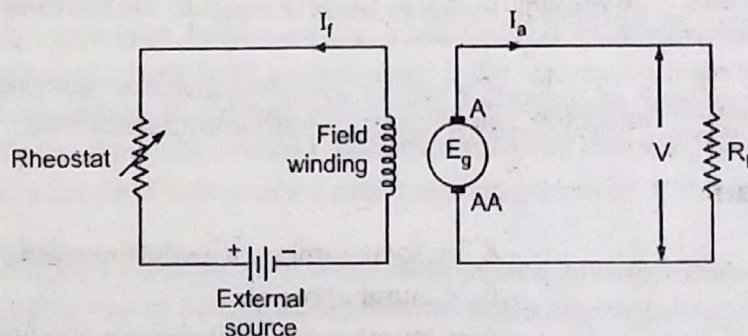


Fig. 9.4. Separately Excited D.C. Generator

(ii) **Self-Excited D.C. Generators** : The field winding is connected to the armature winding and is excited by the current supplied by the generator itself. In such type of dc generators, some flux is always present in the poles due to residual magnetism, which in turn produces a small voltage when the armature rotates. This voltage causes current to flow in the field coils, thus increasing the flux which in turn increases the induced voltage further. The self-excited dc generators are further classified as :

(a) **Series Wound Generator** : Here, the field windings are connected in series with the armature winding. Thus the entire current flows through the field winding and load as shown in figure 9.5. The field winding is designed to have less number of turns of thick wire as it has to carry full-load current. This means that the resistance of field winding is very low.

As shown in the figure above, excitation current is the same as the armature current.

Also, voltage generated, $E_g = V + I_a(R_a + R_f)$

Such generators are not used practically because the generator is excited only when the load circuit is complete. They are however used as boosters.

(b) **Shunt wound Generator** : Here, the field winding has many turns of fine wire and is connected across the armature terminals. Thus the voltage across the field winding is same as the terminal voltage, as shown in figure 9.6

Referring to the above figure, total current

$$I_a = I_{sh} + I_L$$

Shunt field current, $I_{sh} = \frac{V}{R_{sh}}$

Load current, $I_L = \frac{V}{R_L}$

Generated emf, $E_g = V + I_a R_a$

In order to reduce the power loss in the field circuit, the shunt field current should be as small as possible.

Shunt generators find application as charging accumulators and with automatic voltage generators.

(c) **Compound Wound Generator** : It is a combination of shunt and series generator. There are two field windings, one is connected across the armature and the other is connected in series with the armature winding. If the shunt winding is in parallel with both armature and series winding, the generator is called long-shunt generator (figure 9.7a). But if the shunt winding is in parallel with armature alone, it is called short-shunt generator (figure 9.7b).

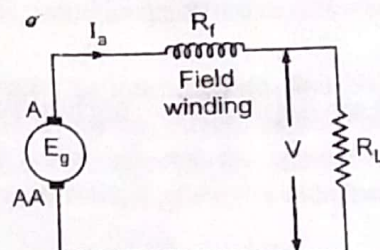


Fig. 9.5. Series Wound Generator

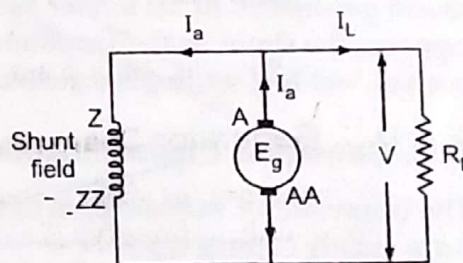


Fig. 9.6. Shunt Wound Generator

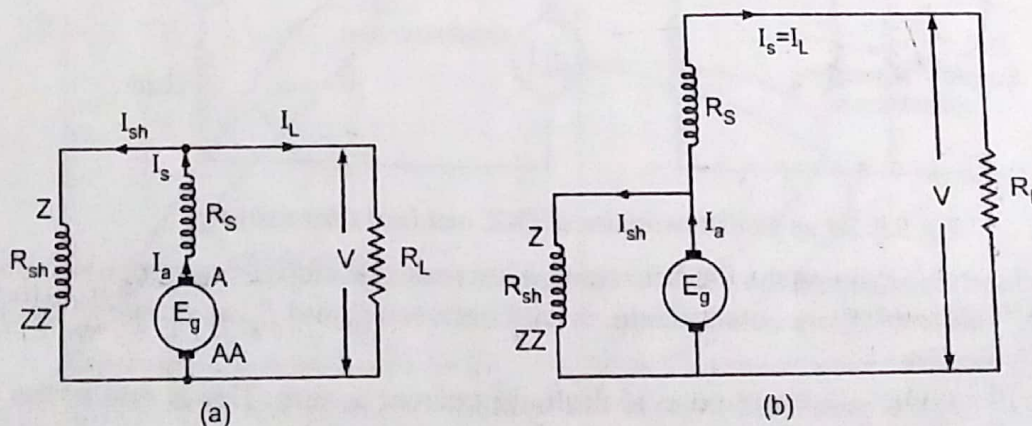


Fig. 9.7. Compound Wound Generator

Referring to figure (a) above,

$$\begin{aligned} I_a &= I_s \\ I_a &= I_L + I_{sh} \\ E_g &= V + I_a R_a + I_s R_s \\ I_{sh} &= \frac{V}{R_{sh}} \end{aligned}$$

From figure (b), we conclude that:

$$\begin{aligned} I_s &= I_L \\ I_{sh} &= \frac{V + I_L R_s}{R_{sh}} \\ I_a &= I_L + I_{sh} \\ E_g &= V + I_a R_a + I_s R_s \end{aligned}$$

Compound generators are also classified as cumulative compound generators and differential compound generators. In the former, series field assists the shunt field, whereas in the latter, series field opposes the shunt field. Compound generators find various commercial applications such as locomotives, welding set, in-plant generation etc.

9.7. D.C. Generator Characteristics

The behaviour of various types of d.c. generator can be studied by their characteristic curves which are mainly of three types:

1. Magnetization characteristic or open-circuit characteristic (OCC) which gives the relation between the induced emf E_g in the generator and field current I_f at no-load with the generator being driven at constant speed.
2. External characteristic that gives the relation between the terminal voltage V and the load current I_L at a given speed.
3. Internal characteristics which shows the relationship between the generated emf E_g at no load and the armature current I_a . We will now discuss these characteristics for different types of d.c. generators.

9.8. Characteristics of Separately Excited D.C. Generators

Figure 9.8 shows the circuit diagram for determination of OCC and load characteristics.

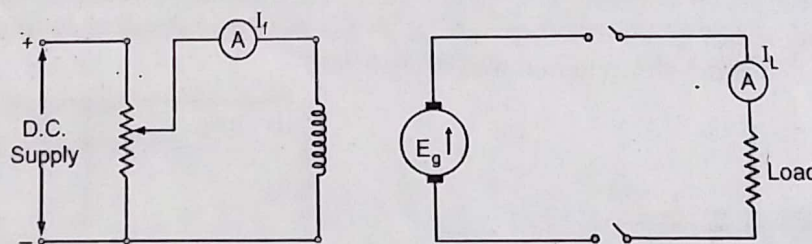


Fig. 9.8. Set up for determination of OCC and Load Characteristics

Keeping the load circuit open, the field current I_f is increased in steps from zero to maximum and the corresponding values of E_g are noted down. A plot between I_f and E_g , as shown in figure 9.9 (a) gives the following results:

- (i) A small voltage is generated even if the field current is zero. This is due to the residual magnetic flux in the poles.

- (ii) The next part of the curve is almost linear till saturation point is reached.
- (iii) The curve now bends as it departs from linearity, and the induced emf becomes almost constant.

For obtaining the load characteristics, the generator is driven at rated speed. The field current is adjusted with the help of rheostat to get rated terminal voltage. Now the load circuit is closed and the load current is increased in steps. The corresponding terminal voltage is noted and plotted as shown in figure 9.9 (b).

As the generator is separately excited, armature current is the same as load current. But the armature reaction will cause a decrease in voltage, thus resulting in a drooping curve; called the internal characteristics. When the ohmic drop in the armature winding with respect to the armature current (represented by a straight line) is deducted from internal characteristic, it gives the external characteristic.

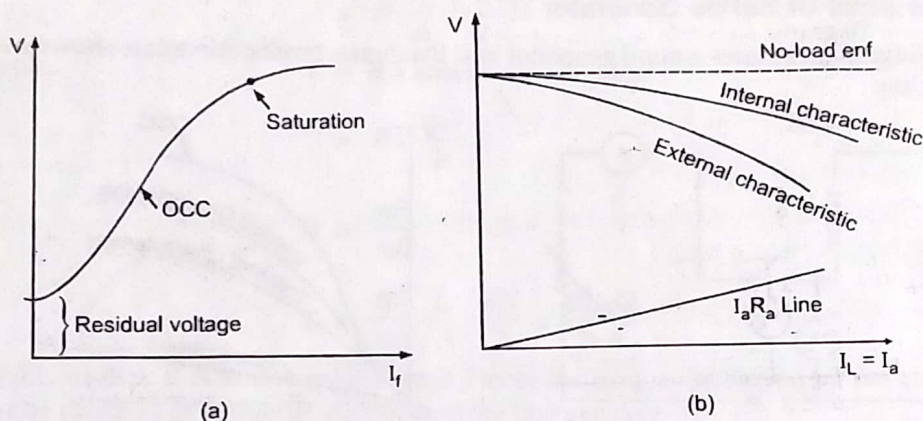


Fig. 9.9. Characteristic Curves

9.9. Characteristics of Shunt Generator

The curve plotted between the generated emf and shunt field current is shown in figure 9.10 (a). When a shunt generator is rotated at constant speed, a small emf is generated due to residual magnetism. This small emf causes a small current in the shunt field winding. The current in turn tends to increase the magnetic flux thereby increasing the generated emf. This cumulative effect is the basis of the building up of emf in shunt generators. For a given magnetization curve, the stable voltage depends upon the total resistance in the field circuit. The slope of the line drawn tangent to the initial portion of the curve (as shown) gives the critical field resistance. If the field circuit resistance is more than the critical value, the generator will fail to excite at the given speed.

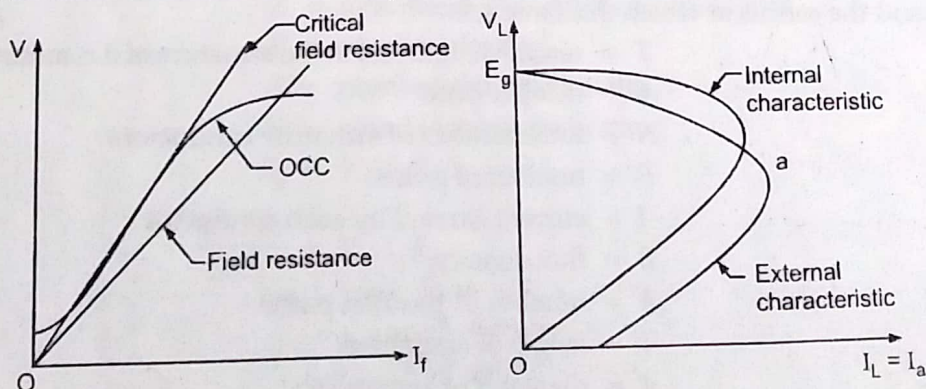


Fig. 9.10. Characteristic Curves of Shunt Generator

The internal and external load characteristics are shown in figure 9.10 (b). In case of shunt generator, the terminal voltage depends on the field resistance and is E_g at no-load. As the generator is loaded, the load current increases causing a decrease in terminal voltage. The effect of decrease in resistance predominates over the effect of decreased terminal voltage till point 'a'. However, if the load resistance is decreased further, the voltage falls at an increasing rate due to the demagnetising effect of armature reaction and voltage drop in armature. Hence, after point 'a', if the load resistance is decreased, the load current will decrease instead of increasing. Terminal voltage will also decrease and the characteristics turn back as shown. The internal characteristic has a similar shape as that of external characteristic. It is above the external characteristic by an amount of $I_a R_a$ and will not intersect the X-axis because of residual magnetism.

9.10. Characteristics of Series Generator

The circuit diagram of a series wound generator and the characteristic curves are shown in figure 9.11 (a) and (b) below :

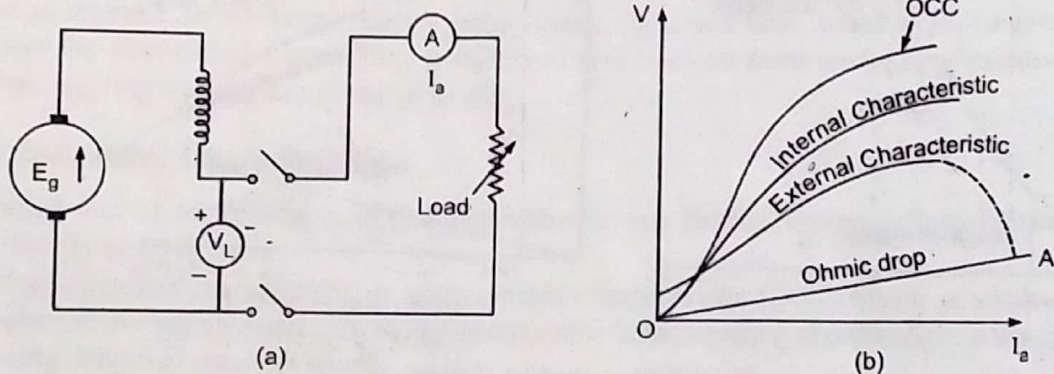


Fig. 9.11. Series Generator and its Characteristics

The magnetic characteristic is same as in the case of shunt generator and is drawn between flux per pole and field current. The internal characteristic gives the relation between generated emf E_g and armature current I_a . The terminal voltage of the generator is equal to the generated emf E_g less the voltage drop in armature and series field respectively i.e., $V = E_g - I_a (R_a + R_s)$. The straight line representing this ohmic drop (OA) is deducted from the internal characteristic to get the external characteristic.

9.11. Torque Developed in D.C. Motor

The measure of causing the rotation of a wheel is called the torque. Torque is measured by the product of force and the radius at which this force acts.

Let

T = torque developed in the armature of d.c. motor

ϕ = flux per pole

N = total number of armature conductors

P = number of poles

I = current carried by each conductor

B = flux density

A = number of parallel paths

l = length of armature

d = diameter of armature.

Now, force acting on a current carrying conductor placed in a magnetic field is given by :

$$F = BIl$$

Torque

$$T = F \times \frac{d}{2}$$

$$= BIl \cdot \frac{d}{2}$$

$$\text{Current in each conductor} = \frac{I_a}{A}$$

∴ Torque developed due to N conductors

$$= B \cdot \frac{I_a}{A} \cdot l \cdot \frac{d}{2} \cdot N$$

Flux per pole,

$$\phi = B \times \text{area under one pole}$$

$$= B \times \frac{\pi dl}{P}$$

$$\Rightarrow B = \frac{\phi P}{\pi dl}$$

$$\therefore T = \frac{\phi P}{\pi dl} \cdot \frac{I_a}{A} \cdot \frac{ldN}{2} = \frac{\phi P I_a N}{2\pi A}$$

In a given d.c. motor, P , N and A are constant. Hence the torque developed by the armature is proportional to the product of armature current and the flux per pole.

i.e.,

$$T \propto \phi I_a$$

For a d.c. shunt motor, flux is constant and hence,

$$T \propto I_a$$

In case of a d.c. series motor, flux is proportional to be armature current, i.e.,

$$I \propto I_a^2$$

9.12. Speed of Motor

As seen in article 9.5, back emf of a d.c motor is given by :

$$E_b = \frac{\phi NPZ}{60A}$$

where

Z = total number of armature conductors

P = number of poles

A = number of parallel paths

N = speed of armature in rpm.

Also,

$$E_b = V - I_a R_a$$

$$\therefore \frac{\phi NPZ}{60A} = V - I_a R_a$$

$$\Rightarrow N = (V - I_a R_a) \frac{60A}{\phi PZ}$$

For a given motor, P , Z and A are fixed.

$$\therefore \text{Speed of motor} \quad N \propto \frac{V - I_a R_a}{\phi}$$

$$\propto \frac{E_b}{\phi}$$

Thus, the speed is directly proportional to back emf and inversely proportional to flux per pole.

9.13. Types of D.C. Motors

The d.c. motors can be classified as :

- (i) Permanent Magnet type
- (ii) Separately excited motor
- (iii) Series wound motor
- (iv) Shunt wound motor
- (v) Compound motor.

9.13.1. Permanent Magnet Motor

It consists of armature and permanent magnets surrounding the armature. Field coils are required only for recharging the magnets.

9.13.2. Separately Excited D.C. Motor

As shown in figure 9.12, the armature and field coils are fed from different supply sources.

Thus, armature current $I_a = \text{Line current } I_L$
 Back emf $E_b = V - I_a R_a$

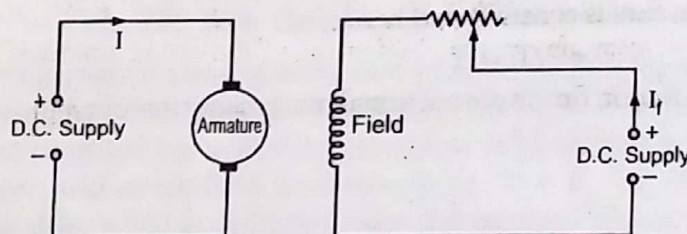


Fig. 9.12. Separately Excited D.C. Motor

9.13.3. Series Wound D.C. Motor

The field coils are connected in series with the armature as shown in figure 9.13

Armature current $I_a = \text{series field current } I_s$
 $= \text{Line current } I_L$
 Back emf developed, $E_b = V - I(R_a + R_s)$

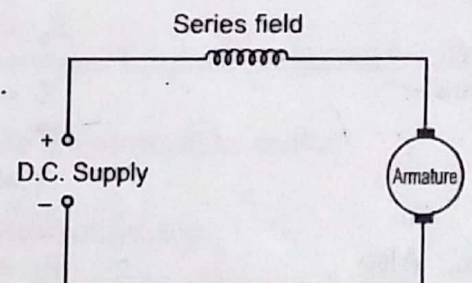


Fig. 9.13. Series Wound D.C. Motor

9.13.4. Shunt Wound D.C. Motor

The field coils are connected in parallel with the armature as illustrated in figure 9.14. The current supplied to the motor is divided into two parts – one through the field winding and the other through the armature.

∴ Input line current,

$$I_L = I_a + I_s$$

where

$$I_s = \frac{V}{R_s}$$

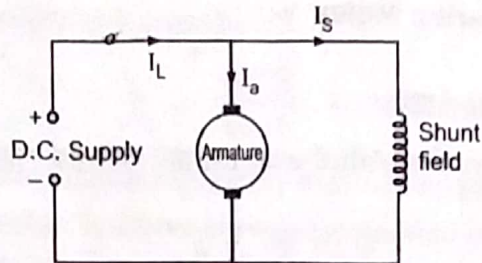


Fig. 9.14. Shunt Wound D.C. Motor

9.13.5. Compound Wound Motor

It has both shunt and series field coils. The shunt field coil has more ampere-turns than the series field coils. Compound motors are further classified as cumulative compound wound and differential compound wound motors.

9.14. Characteristics of D.C. Shunt Motor

9.14.1. Speed-Armature Current Characteristic

In a d.c. shunt motor, armature and field circuits are in parallel. Hence field current I_f does not depend on armature current I_a . If the applied voltage V is constant, field current I_f will remain constant. But practically, due to demagnetizing effect of armature reaction, the net flux is slightly reduced. Now, speed N is directly proportional to back emf E_b or $(V - I_a R_a)$ and inversely proportional to flux ϕ . But since flux ϕ is considered to be constant, so with the increase in armature current, the speed falls slightly as shown in figure 9.15 (a).

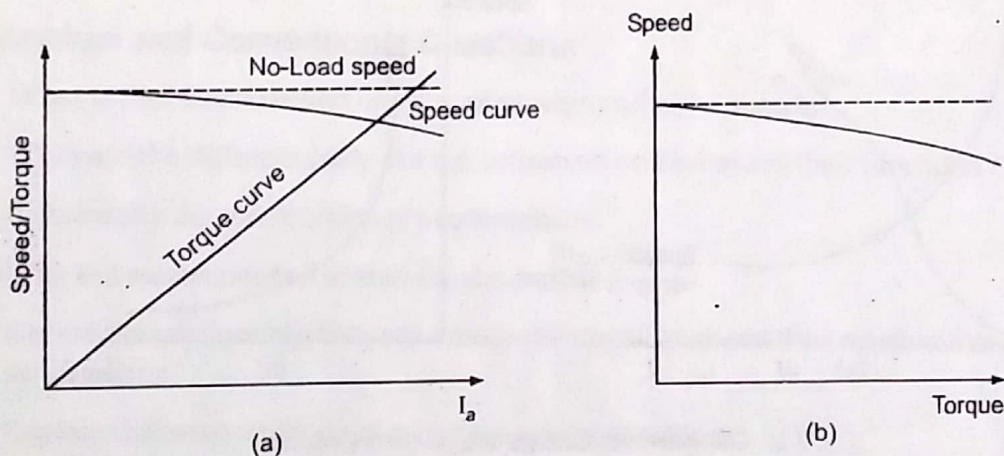


Fig. 9.15. Characteristic Curves of D.C. Shunt Motor

9.14.2. Torque-Armature Current Characteristic

The torque of a d.c. motor is directly proportional to the product of flux and armature current. As the flux is practically constant, the torque rises with the increase in armature current i.e., torque-armature current characteristic is a straight line passing through origin as shown in figure 9.15 (a) above.

9.14.3. Speed-Torque Characteristic

This characteristic curve can be drawn from the above two characteristics and is shown in figure 9.15 (b).

9.15. Characteristics of D.C. Series Motor

9.15.1. Speed-Current Characteristic

In a d.c. series motor, the flux varies with the armature or load current. The speed equation is given by :

$$N \propto \frac{V - I_a R_a}{\phi}$$

$$\propto \frac{V - I_a R_a}{I_a}$$

$$\propto \frac{V}{I_a} - R_a$$

This equation gives a hyperbolic curve as shown in figure 9.16 (a). As the load increases, magnetic saturation occurs and the curve tends to become a straight line. Thus, the series motor is a variable speed motor *i.e.*, the speed varies with the change in line current.

9.15.2. Torque-Current Characteristic

The torque of a d.c. motor is proportional to ϕI_a . But since flux ϕ is proportional to armature current I_a , so $T \propto I_a^2$. Thus, upto the saturation point, the torque-current curve is a parabola as shown in figure 9.16 (a). But after the saturation point, flux ϕ is independent of excitation current and so torque is proportional to I_a . Hence the curve becomes a straight line.

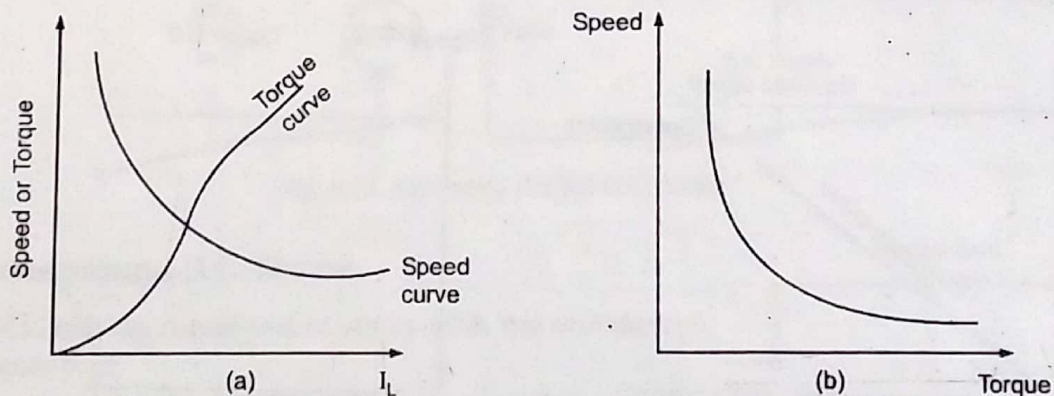


Fig. 9.16. Characteristic Curves of D.C. Series Motor

9.15.3. Speed-Torque Characteristic

It can be drawn with the help of the above two characteristics, as shown in figure 9.16 (b) above.

SALIENT POINTS

- D.C. machine has commutator which converts a.c. into d.c. and vice versa.
- An electrical generator converts mechanical energy into electrical energy.
- EMF equation of a generator is given by

$$E_g = \frac{\phi NPZ}{60 A}$$

- D.C. generators are classified as separately excited and self-excited generators.
- Open circuit characteristic is drawn between generated emf and excitation current.
- Internal characteristic gives the relation between induced emf in armature and armature current.
- External characteristic gives relation between terminal voltage V and load current I .
- The back emf in a d.c. motor,

$$E_b = \frac{\phi NZP}{60 A}$$

- Direction of rotation of a d.c. motor can be reversed by reversing the current through the armature winding or field coils.
- The speed of a series motor increases with the decrease in load. They are best suited for electric traction work.
- Shunt motor is a constant speed motor.
- Shunt motor finds applications requiring medium starting torque such as blowers, fans, conveyors, boring mills etc.

REVIEW QUESTIONS

A. Conceptual and Conventional Questions

1. What is the basic function of a d.c. generator and a d.c. motor?
2. What are the different parts of a d.c. armature and what are their functions?
3. Give briefly the construction of a commutator.
4. Why is a starter needed to start the d.c. motor?
5. Derive the relationship between the torque developed and the armature current in case of a series motor.
6. Explain different methods of excitation of d.c. machines.
7. Explain the nature of open-circuit and external characteristics of a d.c. shunt generator.
8. Derive an expression for the speed of a d.c. motor in terms of back emf and flux per pole.
9. Draw and explain the torque-speed characteristics for different types of d.c. motors.
10. Explain what happens when in a d.c. shunt motor, field terminals are reversed and when armature terminals are reversed.

B. Fill in the blanks with appropriate word/words :

1. In a d.c. machine, iron losses occur in
2. For a given d.c. motor, the speed depends upon
3. The commutator in a d.c. generator is used for changing into
4. In a d.c. machine, the constant losses are and
5. motor gives a high starting torque.
6. The variable losses in a d.c. machine are
7. If a d.c. series motor is started on no-load, its speed will be
8. The rotating part in a d.c. machine is generally called
9. The armature of a d.c. machine is laminated to reduce losses.
10. In d.c. machines, at maximum efficiency, variable losses is constant losses.

Answers :

1. armature, 2. back emf and flux, 3. a.c., d.c., 4. iron loss, mechanical loss, 5. Series, 6. armature copper losses, 7. infinite, 8. armature, 9. eddy current, 10. equal to.

C. Multiple Choice Questions :

1. Thermistors are non-linear resistors having

(a) +ve temperature coefficient	(b) -ve temperature coefficient
(c) +ve or -ve temperature coefficient	(d) none of these.
2. Sensistors are :

(a) +ve temperature coefficient thermistors
(b) -ve temperature coefficient thermistors
(c) Sensitive to very small temperatures
(d) Sensitive to voltage changes.
3. The nature of current flowing in the armature of a d.c. machine is :

(a) d.c.	(b) a.c.
(c) pulsating	(d) none of these.
4. Brushes are provided in a d.c. machine for

(a) preventing sparking	(b) reducing losses
(c) providing a path for flow of current	(d) none of these.
5. The armature of a d.c. machine is laminated to reduce

(a) eddy current losses	(b) hysteresis losses
(c) copper losses	(d) friction losses.

6. The relationship between applied voltage V and back emf E_b for a d.c. motor is given by :
 - (a) $V = E_b - I_a R_a$
 - (b) $V = E_b$
 - (c) $V = E_b + I_a R_a$
 - (d) $V = E_b I_a R_a$
7. If the speed of the prime mover of a d.c. generator is increased, the generated emf :
 - (a) decreases
 - (b) increases
 - (c) remains constant
 - (d) cannot say
8. On loading, the speed of a d.c. shunt motor :
 - (a) increases
 - (b) decreases
 - (c) remains constant
 - (d) decreases sharply.
9. The function of a starter in a d.c. motor is :
 - (a) to control its speed
 - (b) to reduce starting current within limits
 - (c) to avoid sparking
 - (d) to minimize armature reaction effects.
10. The speed of a d.c. shunt motor is to be increased beyond its rated value. This can be achieved by :
 - (a) inserting a resistance in armature circuit
 - (b) inserting a resistance in field circuit
 - (c) applying less voltage
 - (d) applying more voltage.

ANSWERS :

- | | | | | |
|--------|--------|--------|--------|----------|
| 1. (b) | 2. (a) | 3. (b) | 4. (c) | 5. (a) |
| 6. (b) | 7. (b) | 8. (b) | 9. (b) | 10. (b). |



10

Synchronous Machines

OBJECTIVE

After completing this unit, the reader will be able to :

- Understand the operating principle of a synchronous generator and a synchronous motor.
- Classify the synchronous machines and understand the working of essential parts of the machine.
- Derive the emf equation and the expression of the induced emf frequency.
- Understand the behaviour of synchronous motor under varying load.
- Understand the advantages, disadvantages and applications of synchronous motors.

10.1. Introduction

Synchronous machine is a rotating electrical machine in which excitation is required for both stator and rotor. A synchronous machine is widely used in electrical power system. Synchronous generators, also called alternators, along with the steam turbine are termed as turbo-generators. Thus, synchronous machines are generally constructed in large sizes. The point worth remembering is that in synchronous machine, torque is produced only when the machine is running at synchronous speed. Since it is an a.c. machine, no rotating commutator is required.

10.2. Operating Principle

Fundamentally, the operating principle of a synchronous machine is same as that of a dc machine. The only difference lies in the fact that a synchronous machine is an ac machine and thus there is no need to rectify ac into dc. Hence, a commutator is not required. The normal speed of the rotor or the speed at which it produces rated frequency is called synchronous speed of machine. For this reason, these machines are called synchronous machines.

10.3. Classification of Synchronous Machines

According to their applications, synchronous machines can be classified as synchronous generators and synchronous motors. A synchronous generator receives mechanical energy from a prime mover and delivers electrical energy. A synchronous motor receives electrical energy from ac mains and drives mechanical load.

Synchronous machines can be single phase, two-phase or three-phase type. Being large capacity machines, mostly three-phase machines are used.

Depending upon the type of construction of rotor, synchronous machines are classified as salient pole machines and cylindrical rotor machines.

Based on the prime movers to which the machines are mechanically coupled, they are also classified as turbo-generators, diesel engine driven generators and hydro-generators.

An important classification of synchronous machines is as rotating armature type and rotating field type. Practical considerations of design are met by a rotating field structure and a stationary armature. The advantages of stationary armature are :

- (i) Stationary armature is not subjected to mechanical stress. Also more space is available on the stator which is outside the rotor. This, it is easier to insulate stationary armature windings.
- (ii) The armature winding can be cooled easily due to large surface area.
- (iii) The load terminals can be connected directly with the stator as no slip-rings and brushes are required.
- (iv) Only two slip-rings are required for dc supply to rotor.
- (v) Higher speed of rotor is possible.

10.4. Construction

Synchronous machine consists of two main parts—armature or stator, and field magnet system or rotor.

1. **Stator:** The stator core or armature is a stack of cylindrical steel laminations, having slots on their inner periphery to accommodate armature conductors. It is housed in a cast iron frame as shown in figure 10.1. To minimize the hysteresis and eddy current losses, silicon steel laminations are used, which are insulated from each other with paper or varnish. Forced ventilation or hydrogen is used as a cooling medium.

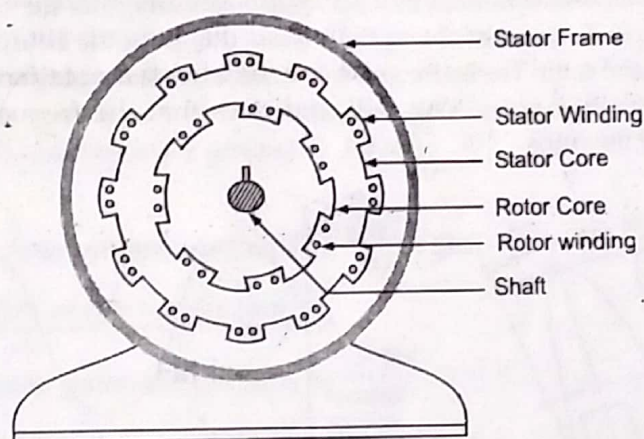


Fig. 10.1. Construction of Synchronous Machine

2. **Rotor:** The rotating part of the machine gets the exciting current through two slip rings and brushes. Based on the structure, rotor is of two types—salient pole type and cylindrical or non-salient pole type.
 - (i) **Salient Pole Type:** This type of rotor is for low speed alternators. The salient poles are made of thick steel laminations, projecting out from the rotor core as shown in figure 10.2. The air-gap between stator and rotor is non-uniform, increasing from the pole centre to pole tips. Thus, the distribution of flux over the armature is sinusoidal, resulting in a sinusoidal generated emf. The salient pole structure has the advantage of large diameter and short axial length.
 - (ii) **Cylindrical Pole Type:** This rotor is used in high speed alternators. The air gap between stator and rotor is uniform throughout, and the diameter of rotor is reduced with increased axial length. The dc field windings are embedded in rotor slots and are of distributed type. The cylindrical pole structure, which consists of cylindrical steel forging with radial slots has less windage loss and gives noiseless operation.

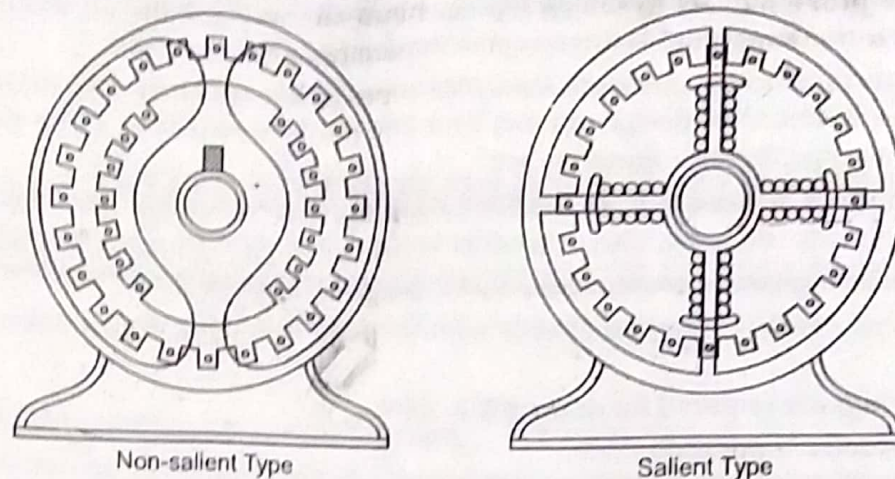


Fig. 10.2. Types of Rotor

10.5. Synchronous Generator

A synchronous generator or alternator has the armature winding on stator and field winding on rotor. The field winding is excited by d.c. as shown in figure 10.3. When the rotor is rotated by means of some prime-mover, flux is produced. The armature conductors cut this magnetic flux. As per Faraday's law of electromagnetic induction, an emf is induced in the armature conductors. The polarity of this induced emf is to cause a current to flow that opposes any change in the flux linkage of the stator coil. This stator current produces a revolving field that cuts the rotor field, producing a flux linkage common to both stator and rotor. The stator and rotor fields are stationary with respect to each other, and have the same number of poles. This indicates that the frequency of induced emf is synchronized with the speed of the rotor.

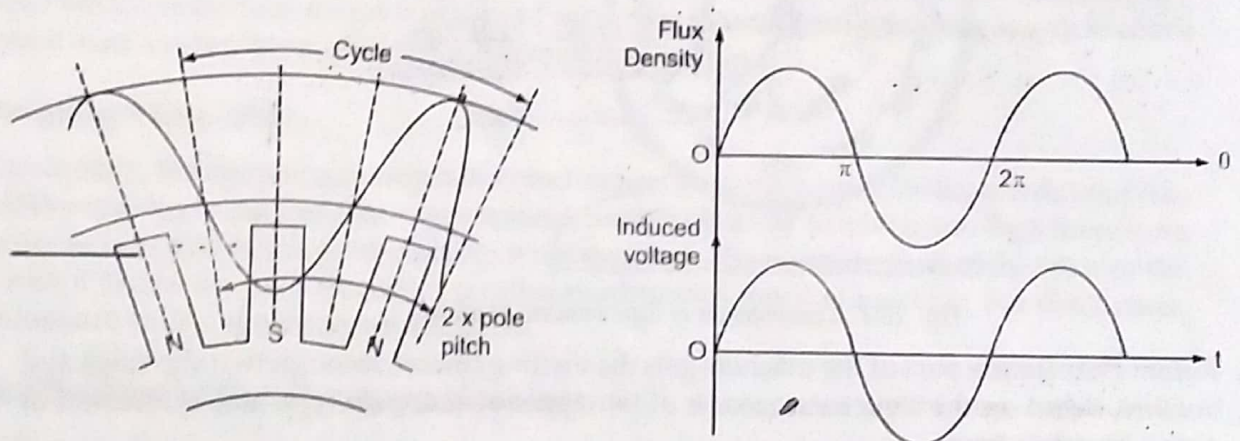


Fig. 10.3. Synchronous generator

10.6. Induced EMF Frequency

As shown above in figure 10.3., emf induced in the conductor completes one cycle in an angular distance equal to twice the pole pitch.

Let

P = number of poles on rotor.

n = number of revolutions made per second by rotor

Then, number of cycles of emf completed in one revolution

$$= P/2$$

∴ Number of cycles per second = Frequency f

= Number of emf cycles per revolution $\times$ number of revolutions of rotor per second

$$\text{i.e. } f = \frac{P}{2} \times n$$

$$\Rightarrow f = \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120}$$

where N is the number of revolutions made by rotor per minute.

Thus, the frequency of induced emf is directly proportional to the number of poles and speed of rotor.

10.7. EMF Equation

Let

P = total number of poles

ϕ = flux per pole in weber

N = speed of rotor in rpm

T = total number of turns per phase

K_d = distribution factor

K_p = pitch factor

(The armature winding is distributed with fractional pitch coils and some armature slots are left empty leading to distribution factor K_d).

Flux cut by each conductor per revolution = $P\phi$

Flux cut by each conductor per second = $P\phi \times \frac{N}{60}$

$$\therefore \text{Average generated emf} = \frac{P\phi N}{60} \text{ volts.}$$

Now one turn consists of two conductors.

$$\therefore \text{Average emf generated per turn} = \frac{2P\phi N}{60}$$

$$\therefore \text{Average emf generated per phase} = \frac{2P\phi N}{60} \times T \text{ volts}$$

$$\text{Now, frequency } f = \frac{PN}{120}$$

$$\therefore \text{Average emf per phase} = 4f\phi T$$

$$\text{or, rms emf generated per phase, } E = 1.11 \times 4f\phi T = 4.44 f\phi T \text{ volts.}$$

If K_d and K_p are taken into consideration, then

$$E = 4.44 K_d K_p f\phi T$$

10.8. Synchronous Motor

In a synchronous motor, the field is energized by direct current, and the armature winding is connected to a 3-phase ac supply. Essentially, the construction of a polyphase synchronous motor is

same as that of an alternator. The main parts of a 3-phase synchronous motor includes : (i) Stator core with armature winding, (ii) rotating field with damper winding and slip rings, (iii) brushes, and (iv) bearings shields. While the field created by the stator is rotating, the field of the rotor is stationary. As equal and opposite torques are produced by the rotating field, the rotor does not move and the synchronous motor has no starting torque. The d.c. excited rotor is hence brought to synchronous speed by an auxiliary device.

The stator core and windings of a synchronous motor are similar to those of a 3-phase alternator. In addition to the exciting winding, it also is provided with squirrel cage winding on the rotor, whose slots are parallel to the shaft.

10.9. Principle of Operation of Synchronous Motor

Consider a two pole synchronous motor as shown in figure 10.4. The rotor has two poles and stator has two poles per phase. When a 3- ϕ supply is given to stator, a rotating magnetic field is setup in the stator, which rotates at synchronous speed. Say, at any instant of time, stator poles are in the N-S direction and the rotor is as shown in the figure.

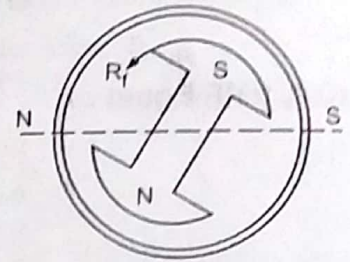


Fig. 10.4. Synchronous motor

Because of the repulsion between like poles, the rotor experiences a force in the anticlockwise direction. After sometime, the stator poles which are rotating at synchronous speed will reverse their positions. Now, the rotor will experience an attractive force, and this torque tries to move the rotor in clockwise direction. The net result is a stationary rotor. Hence, a synchronous motor has no self starting torque. To start this motor, it is rotated at a speed equal to the synchronous speed before energising its field winding. There are two methods of starting a synchronous motor :

- (i) **Auxiliary Starting :** An auxiliary motor is mechanically coupled to the synchronous motor. The function of this motor, which can be an induction motor or d.c. motor, is to bring the synchronous motor to its synchronous speed.
- (ii) **Induction Motor Starting :** In this method, damper windings are embedded in the rotor slots. When the speed of the synchronous motor reaches almost the synchronous speed, the field windings are excited using a d.c. source.

10.10. Effect of Load on Synchronous Motor

As the synchronous motor has no self-starting torque, it is driven by some external means, as discussed in section 10.9, and is brought to the synchronous speed. Now the stator and rotor fields are at the same speed, i.e., the centre lines of the stator and rotor poles coincide as shown in figure 10.5(a). At this time, the induced or back emf E_b is equal to the applied voltage, but opposite in direction. Under these conditions, the motor is said to be floating.

After synchronizing with the mains, the rotor drive is removed. Now there is no change in speed, but only a shift in the relative positions of the two poles. The rotor pole falls back a little and the angular displacement between the rotor and stator poles is called the torque angle (δ), as shown in figure 10.5(b). The applied voltage V and the induced back emf E_b are no longer in direct opposition.

The resultant voltage E_R of V and E_b causes a current $I_0 \left(= \frac{E_R}{Z} \right)$ to flow in the stator winding. Under this condition of no-load, the power drawn by the motor is $VI_0 \cos \phi_0$ from the mains which keeps the motor running at synchronous speed.

Finally, when the motor is loaded, the rotor pole line falls back a little more as compared to the

stator pole. This is depicted in figure 10.5(c). The torque angle increases, thereby increasing the resultant voltage E_R . It implies that the current drawn from the mains will increase. But if the motor is overloaded, the rotor pulls-out of synchronism and the motor comes to a standstill.

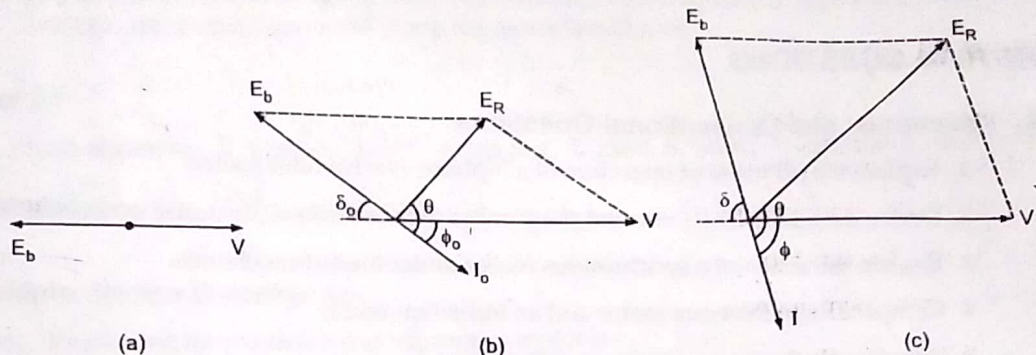


Fig. 10.5. Effect of load on synchronous motor.

10.11. Applications of Synchronous Motors

Synchronous motors require external dc excitation source. Also, the cost per kW output is more than that of an induction motor. Hence, synchronous motors are rarely used below 40 kW output. They find application in power factor correction, voltage regulation and constant-speed-constant-load drives. Low speed synchronous motors are used as vacuum pumps, line shafts, dc generators, rolling mills, centrifugal pumps etc. Synchronous motors in huge sizes are employed to drive rotary kilns and crushers, where they operate at very low speeds.

SALIENT POINTS

- A synchronous machine is an ac machine that requires excitation for both stator and rotor.
- Synchronous generator or alternator converts mechanical power into electrical power at a desired frequency.
- The speed of the rotor at which it produces rated frequency is called synchronous speed.
- Although a synchronous machine can be of rotating armature type or rotating field type, but practically rotating field structure and stationary armature are used.
- The rotor of a synchronous machine can be of salient pole type or smooth cylindrical type.
- The frequency of the induced emf is given by

$$f = \frac{PN}{120} \text{ Hz}$$

where P is number of poles on rotor and N is the speed of rotor in rpm.

- The rms value of the emf generated per phase is given by :

$$E = 4.44 f \phi T \text{ volts.}$$

- The construction of a synchronous motor is same as that of an alternator.
- A synchronous motor has no self-starting torque. It requires some external source to drive it to the synchronous speed.

- Synchronous motors are economical for low speeds and high kW output applications.
- Synchronous motors are used for power factor correction, voltage regulation and constant speed-constant load drives.

REVIEW QUESTIONS

A. Conceptual and Conventional Questions

1. Explain the principle of operation of a 3-phase synchronous motor.
2. Derive the relationship between the number of poles, frequency and speed of alternator.
3. Explain the action of a synchronous motor under loaded conditions.
4. Compare a synchronous motor and an induction motor.
5. Why is a synchronous motor not self-starting?
6. What is the purpose of damper windings in a synchronous motor?
7. Discuss the industrial applications of synchronous motors.
8. What are the different methods of starting a synchronous motor?
9. Explain what happens when a synchronous motor is loaded beyond its pull-out torque.
10. Answer the following briefly :
 - (i) What is a rotating magnetic field?
 - (ii) At what voltage is the field of an alternator usually excited?
 - (iii) How can the direction of stator revolving field be reversed?
 - (iv) How does the power factor of a synchronous motor working on constant load, depends upon its excitation?
 - (v) Why is salient pole rotor not suitable for high speed alternators?

B. Fill in the blanks with appropriate word/words :

1. The direction of rotation of a synchronous motor can be reversed by reversing the supply
2. The angle between the rotor poles and stator flux is called angle.
3. The maximum value of torque angle in a synchronous motor is
4. The power factor is when the field of a synchronous motor is under excited.
5. A synchronous motor has self-starting torque.
6. Synchronous motors are costly and complicated than induction motors.
7. A synchronous motor is identical in construction to an and converts ac electrical power into power.
8. The frequency of induced emf in the stator of a synchronous machine depends upon the number of and of rotor.

9. To obtain a continuous torque, the rotor should rotate at a synchronous speed given by the expression
10. When a synchronous motor is on no-load, the back emf is equal and opposite to the applied voltage, and under these conditions, the motor is said to be

Answers :

1. phase sequence, 2. torque, 3. 90° , 4. lagging, 5. Zero, 6. more, 7. alternator, mechanical,
8. poles, speed, 9. $N = \frac{120f}{p}$, 10. floating.

C. Multiple Choice Questions :

1. The starting torque in a 3- ϕ synchronous motor is
 - (a) high
 - (b) low
 - (c) very low
 - (d) zero
2. A slow speed alternator is known as :
 - (a) turbogenerator
 - (b) non-salient pole alternator
 - (c) water wheel generator
 - (d) none of these
3. The regulation of a three-phase alternator is given by :
 - (a) $\frac{E-V}{V} \times 100\%$
 - (b) $\frac{V-E}{E} \times 100\%$
 - (c) $\frac{E-V}{E} \times 100\%$
 - (d) none of these.
4. The most commonly used prime mover for slow speed alternators is :
 - (a) diesel engine
 - (b) hydraulic turbine
 - (c) petrol engine
 - (d) steam turbine
5. A 3- ϕ synchronous motor which is provided with a damper winding can be started as a
 - (a) 3- ϕ synchronous motor
 - (b) 3- ϕ induction motor
 - (c) 3- ϕ alternator
 - (d) 1- ϕ induction motor
6. In a synchronous motor, the magnitude of back emf E_b depends on
 - (a) load on the motor
 - (b) speed of the motor
 - (c) d.c. excitation
 - (d) both speed and flux
7. If the field of a synchronous motor is under-excited the power factor will be :
 - (a) lagging
 - (b) leading
 - (c) unity
 - (d) more than unity

8. An electric motor in which both stator and rotor fields rotate at the same speed is called :
(a) dc motor (b) universal motor
(c) synchronous motor (d) none of these
9. A synchronous machine is called a doubly-excited machine because
(a) it can be overexcited (b) it needs twice the normal exciting current
(c) both rotor and stator are excited (d) it has two sets of rotor poles
10. When load on a synchronous motor is increased, its armature current is increased provided it is :
(a) over excited (b) under excited
(c) normally excited (d) all of the above.

Answers :

- | | | | | |
|--------|--------|--------|--------|----------|
| 1. (d) | 2. (c) | 3. (a) | 4. (b) | 5. (b) |
| 6. (c) | 7. (a) | 8. (c) | 9. (c) | 10. (d). |

